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INFORMATION ALTRUISM

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by

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Abstract

The research follows the dissemination of an innovative educational technology solution created by the Transformational Learning Services (TLS) group. The technology, referred to as YourNetwork, is built on Drupal, a Free Open Source Software (FOSS) Content Management System (CMS). This research explores the social movements and decisions that lead to Drupal and YourNetwork's dissemination at a large university referred to as CentralU. This has implications beyond just open source software in large organizations, into methods that can be employed by any social movement that leverages information technology.

This research study reinforces politically imbued technological artifacts are just as important as the solutions they provide. It reviews the social and technological activism required to help foster adoption of innovation. The research suggests that the giving away of resources, time, energy and support associated with code transforms the nature of the relationship between actants in a previously stable Actor Network. The concept identified is a new strain of activism the research refers to as Information Altruism. Information Altruism extends the methods available to organizational activists who seek to change their culture.

Information Altruism lays out factors that must be juxtaposed to the current organizational culture in order to be applied. These factors include an established currency, power flow, value system and the ability to plant a flag in the ground for your new cause. These concepts are explored through analysis of interviews with members of CentralU after efforts have been donated across the organization. Observational data and email communications also help to back the argument that a small group of Drupal users were able to transform the perception of where technology should originate.

Table of Contents

List of Figures vi

Acknowledgments vii

Abbreviations viii

Glossary ix

- New Terms ix

Special Note x

Problem Statement 1

- Why this research, why now 3
- Specialized knowledge 5

Background 7

- Today 9

Literature Review 11

- Diffusion of Innovations 11
- Open Source 13
- Organizational Technology Decision Making 18
- Activism 20

Theory 25

- Technology movements and the politics of free / open source software 25
- Do Artifacts have Politics? 25
- Actor-Network Theory 26
- Cross-section 28

Methods 32

- Duration 32
- Autoethnography 32
- Participant Observer 33
- Research Questions 34
- Interviews 35
- Email / Message textual analysis 36
- Technological Agency 37

Data 38

- Approachability of technology 41
- Anti-establishment perception 44
- Groupthink 47
- Independence and empowerment 52

The wearing of "Many Hats" 55
Diffusion of the innovation 56
Interview Conclusions 59

Discussion 62

Information Altruism 62
Factors 63
Other Examples 65
Theory Contributions 67
Knowledge Contributions 69
Information Altruism for Organizational Activism 71
Information Altruism for Managers 72
Information Altruism for Governments 73
Information Altruism for IT organizations 73
Information Altruism for Activists 76
Future research 78

Bibliography 79

Appendix A: Interview Questions 82

Appendix B: Interview demographic 84

Appendix C: Conversation Data 87

Appendix D: IRB Approval Form 88

List of Figures

Figure 1 Diffusion of Innovations Model (Rogers, 2005, p. 12) 12
Figure 2 Key drivers of open-source adoption in U.S. Higher Education (van Rooji, 2009, p. 692) 15
Figure 3: A Framework to Investigate Open Source System Adoption (Glynn, 2005, p. 226) 17
Figure 4 DevOps Intersection visual (Rajiv.Pant, 2012) 29
Figure 5 DevOps Workflow 30
Figure 6: Diffusion of Innovation within CentralU 57
Figure 7: Innovation diffusion based on when people learned about system 58
Figure 8 Diffusion of Innovation with Technologist Zero's influence 60
Figure 9 Interaction Flow non-push based email 68
Figure 10 Interaction flow after push based email change 68

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To my children: I hope one day you see the world poppa has tried to set in motion for you.

To the Drupal community: The future goes to those who build our collective tomorrow. Never stop encouraging the free flow of information. A better, more open and transparent world is in our reach if we but allow it to be given away.

Never stop innovating
Ex Uno Plures
btopro

Abbreviations

- ANT – Actor-Network Theory
- CMS – Content Management System
- CSCMS – Centrally Supported Content Management System
- CSLMS – Centrally Supported Learning Management System
- DevOps – Developer Operations
- FOSS – Free Open Source Software
- LMS – Learning Management System
- TLS - Transformational Learning Services
- TZ – Technologist Zero

Glossary

DevOps

Developer Operations, the structured command sets that a systems administrator plug into a command prompt in order to have the machine execute requests. DevOps is different from traditional system administration in that it leaves placeholders for the input and effectively captures the commands entered into larger, more complex command sets that require less effort to enact more places.

Drupal

Drupal is an open source Content Management System (CMS) that powers approximately 2% of the web. It is a popular framework for the construction of complex, high scale applications. It is also popular among activist groups as many companies in the community were born of the 2004 Howard Dean campaign.

New Terms

Information Altruism

The donation of effort changes the perceived value of work, effectively devaluing current works. The manipulation of the currency of one's efforts allows for radical changes within organizations.

Information Altruist

Someone who puts into practice the donation of efforts in order to modify their institution.

Special Note

The following is an Autoethnographic research study, which utilizes pseudonyms in order to protect the organizations, technologies and interview respondents involved. This research takes place at a real university, which is referenced as **CentralU** for the purposes of this paper. Key technologies and organizational units have also been renamed to things like **Centrally Supported Content Management System** (CSCMS), **Centrally Supported Learning Management System** (CSLMS), **YourNetwork Platform** and **Transformational Learning Services Group** (TLS).

The author is referred to throughout the research both in the first and third person through the pseudonym **Technologist Zero (TZ)**. This is because I effectively served two roles in the research. There's the work I did that people would know about, where I helped communicate and spread adoption of the YourNetwork platform directly. Any time I was building community in the larger ecosystem through the donation of efforts, I refer to myself as Technologist Zero.

The motivations of Technologist Zero are to expand adoption of Drupal at CentralU in order to establish a new community minded culture. By building this community, the idea is that YourNetwork would be able to be adopted more easily. This distinction is important as we explore interview responses associated with comfort in adopting an untraditional community platform.

Problem Statement

Colleges and universities have been fed a false narrative. Select a Learning Management System (LMS) from one of only a few vendors, or go it alone and risk failure. Institutions are expected to focus their efforts on research efforts and leave product development to professionals.

This narrative is dated and maligns our institutions, which house many of the world's greatest intellectual minds. They are home to faculty that light up the minds of the next generation of scholars, politicians, teachers, philosophers, engineers, artists, technologists and scientists. They teach in ways that connect with their target audience; and yet, through tradition, are expected to educate all through the same technology platforms as any other subject.

The LMS market on the other hand is just that, a market. The incentive is not to capture the innovative spirit of educators but instead to capture market share. Education is faced today, as it has been for far too long, with online educational environments designed by the boardroom, not by the classroom.

Times have changed in part due to notions such as Moore's Law (1965), techniques known as DevOps and the culture of Free Open Source Software (FOSS). Individuals can now build in minutes what used to take large corporations months or years. They can deploy architecture with 100% accuracy, moments after automated tests indicate the build will be successful. They no longer need rooms of people dictating how they teach; they can control and craft their own solutions with minimized risk.

In the late 90's and early 2000's, decisions to purchase large-scale LMS solutions were driven primarily by market pressures. Universities needed to adopt technologies from vendors in order to remain cutting edge. This legacy is no longer possible. In order to remain state of the art, institutions need to produce innovative

solutions rather than look for them on a shelf. In order to remain cutting edge, the solution is not to follow the crowd, but to chart a course offered by collaborative, open software development.

This is one of the largest problems in education today; how to provide innovative, low cost, engaging course experiences, while avoiding risk of negatively impacting learning outcomes.

Does education continue to go the way it always has in educational technology? Or does it introduce risk and adopt radically new approaches to pedagogy and system development in order to provide learners with the best experience possible. How does technology align with the shape of institutions in order to scale those solutions socially?

Because the pace of advancement in information technology is too great to pin down, this research focuses more on the philosophical and political implications of breaking away from business as usual and charting a new course. The research explores how innovation can diffuse throughout organizations that are locked into such legacy mindsets, as well as explores tactics that can be applied to any organization or social movement.

The research study focuses on a large land grant university called CentralU. It explores how locally grown open source solutions spread through an organization and how social engineering is essential to fostering dissemination. The research tracks how distributed technology solutions can maintain support through an organic community structure as opposed to traditionally centralized management. It also explores techniques that can be utilized in order to make organizations more receptive to new forms of information technology implementation.

The research reviews a technology solution that heavily leverages automation and industry standard open technologies in order to collapse team sizes, while improving outcomes. Not only do these technologies replace the need for traditional

vendor solutions, they enhance the relationship of vendors with the institution as they are equal partners in building collaborative solutions, rather than selling one size fits none products. These technologies more importantly increase institutional efficiency, improve perception of learning technology by faculty and staff, and ultimately lead to more positive learner outcomes.

The research follows the dissemination of an innovative educational technology solution created by the Transformational Learning Services (TLS) group. The technology, referred to as YourNetwork, is built on an open source Content Management System (CMS) called Drupal. This research explores the social movements and decisions that lead to YourNetwork being disseminated across CentralU and beyond. This has implications that reach beyond just open source software in large organizations, into the structure and methods of successful social movements.

How can we prepare organizations for radical shifts in technology implementation when notions of support, community, control and authority are shifting to decentralized models via the Internet and Internet of Things? The driving questions that this research seeks to address are:

1. Can we make CentralU more accepting of the Drupal CMS by fostering a community of practice?
2. Can we supplant community through the donation of effort?
3. If information is power, how does the donation of information affect power flow and authority?

Why this research, why now

The researcher is uniquely positioned to engage in this research because of access to active maintainers of the YourNetwork technology solution. The research follows the technology from shortly before diffusion into the organization and tracks how and to whom the innovation disseminated. Having access to YourNetwork developers and founders has enabled the researcher to gain unprecedented access

to the mindset of those producing and distributing the platform, beyond what any website or documentation could reveal.

The YourNetwork platform is also starting to be adopted by other universities and in vendor communities. It is spreading far beyond its original point of creation and diffusing into the larger educational complex. For this reason, it is important to study and document how it spread within CentralU. By understanding how this innovation was produced, more innovations like YourNetwork can potentially be propagated into a stagnant, vendor driven educational technology (edtech) sector. With an increasing reliance on online course technology to deliver social experiences to students, this research is becoming increasingly important to understand.

Society is also at a unique time in the development of technology, particularly open source applications. The pace of technology production is exponentially accelerating. Developers are now creating frameworks that author code, have it tested by quality assurance robots and notify other scripted workflows to do the job of many developers, automatically. This automation of the automator, is leading to exponential gains in the creation, scale and capabilities of new technologies.

Because of this, it is impossible to fully vet products before they are already outdated by newer, better solutions. This has lead to many institutions remaining at a standstill; unable to review new products and adopt them before being immediately dated.

The landscape is confounded by additional societal factors influencing education. Reductions in government funding at the state and federal level, combined with outdated regulations and purchasing practices, have made it impossible for institutions to react and shift quickly to new trends. This problem can also be seen in local and federal governments with lack of ability to get out of lengthy contracts for legacy vendor solutions.

In both instances open source and local, collaboratively developed, solutions have been seen as the way to remedy this situation. It is imperative that we explore the role Free Open Source Software (FOSS) plays in alleviating regulatory, social, and pace of change driven constraints. Open source means that institutions of all kinds no longer have to engage in traditional purchasing practices. Nor do they need to get caught in feedback loops of solutions that cannot be fully vetted or implemented.

Specialized knowledge

I am uniquely positioned to study the YourNetwork platform because of specialized knowledge in the field. I am a web developer turned learning technologist who has been working with instructional designers for over a decade. As a result, I am able to understand both technological and educational sides of edtech. This allows me to both assess and create educational technology. This background knowledge gives me unique insight into both the state of education and technological solutions.

The YourNetwork founding team serves on many committees at CentralU. This has connected the research with other adopters of the YourNetwork platform for the interview portion of my data collection. These relationships made it easier to solicit respondents for interview with greater accuracy as to their involvement and knowledge of the platform.

My software development background enables me to both assess and produce complex technology solutions for education. Whether development of new programming techniques, site building, system administration or theme development; I am well versed in all aspects of development. This enabled me to conduct more engaging interviews with technically oriented respondents that connected with their role in the organization.

I also have an extensive history in developing against the Drupal CMS and have contributed hundreds of projects to the Drupal ecosystem. I am a leading voice in the Drupal in Education community and speak extensively on the topic at industry events. I speak extensively on using FOSS to change organizations, Drupal best practices and spreading the labor of love that is the philosophy of open source. YourNetwork as a technology solution has also inspired the development of other Drupal based edtech systems in industry, and parts of the platform are actively used in other Drupal LMSs.

Just as important as access to the founding members of YourNetwork, is having access to CentralU's Drupal support community. This network of people was able to further promote the YourNetwork platform and recruit other developers from outside of edtech to develop against it.

By building and understanding how a non-education focused CMS could be used by education, the founders effectively inherited a wealth of knowledge and developer base that otherwise would not exist. This lead to the perception of a much larger community at times than actually would have existed in a project of this architecture and point of origin. It is important to note that YourNetwork members are but a subset of the larger Drupal community.

Background

CentralU is similar to other large-scale higher education institutions. It is made up of many colleges, units, and departments all providing services in support of students or the institution itself. These units clash subtly due to the size of the institution, as there is always overlap in service coverage at various levels. CentralU has many traditional, top-down style IT implementations. These typically make sense in the support units but many colleges reject these solutions, regardless of what they are, in favor of locally managed ones.

The culture of CentralU is one of autonomy of the colleges, with central support units unable to relate to desires for autonomy. The two forces, those of sustainability and control versus academic freedom and flexibility, often clash in areas of educational technology delivery. Failure of any project, whether legitimate or not, spread Fear, Uncertainty and Doubt (FUD). Any concerns that CentralU support units could not adequately meet needs in a solution space would lead to a further fragmenting of solutions across the various colleges.

Two particular perceived software failings play an important role as part of the backstory to the YourNetwork research study. These come in the forms of perceived limitations in the Centrally Supported LMS (CSLMS), and policy based limitations of the Centrally Supported CMS (CSCMS).

Both CSLMS and CSCMS were traditional, top-down IT implementations. Both of these decisions were products of what was the popular methodology of implementation in the early 2000s. LMS technology just started to stabilize and courses began being augmented via online learning technology around the late '90s and early 2000s. During the time of this research, 2007 – 2015, CSLMS was utilized and was the only centrally supported solution.

CSCMS was adopted around 2005 as a centrally supported CMS. This was a mix of top-down and enforced local support structure. CSCMS was built on an open source platform that was popular at the time of selection and had a dedicated developer community. The community at the time of selection was not much dissimilar from the global Drupal community. The key difference between these CMS communities was that they were built on different underlying technologies, meaning their developer communities rarely overlapped.

CSCMS had a unique support structure in that colleges and units that adopted CSCMS were required to pay a support fee to the central unit. This granted them access to additional support from a centrally supported, but locally funded, team of developers and project managers. Local adopting units were also required to hire a local support point person to build and manage their group's sites in the CMS, as per the terms of the agreement when adopting CSCMS.

Drupal lacked formal central support during the time of the research. Initially, those that adopted Drupal were met with intense political pressure. As a result, the CSCMS and Drupal communities, while both open source and working for the same institution, were adversarial in nature. This lead to much FUD spread between those supporting either project as they wrestled for control of the technological kingdom.

This may seem like a strange thing for developers to squabble over, the underlying technology used to do their job, but it is very common. Developers train for years to become experts in their field and a large part of that involves knowing how to develop against a specific configuration. Any organizational movement away from your technology configuration means a move away from your skillset. This movement or perception of movement away from that which you've worked years at mastering can be very threatening to developer communities.

This is the context in which the YourNetwork platform was born. Created by a small college outfit known as the Transformational Learning Services (TLS), it was outside the support structure of the CSCMS. It is a platform built on a non-traditional LMS technology, the foundation of which had no central support and was actively campaigned against both publicly and privately. This left it at times in direct opposition to the centrally supported CSLMS as well.

Still, despite downward pressures from multiple central organizations, the YourNetwork innovation managed to diffuse into the organization. Despite no top-down support, the research will focus on how the grassroots found a way to diffuse the innovation throughout parts of the organization.

Today

CSCMS has effectively been disbanded. There is no longer central support for this platform and its support community has effectively left the institution. This is due to a variety of factors but primarily the diffusion of Drupal and Drupal by proxy in the form of YourNetwork. All new CMS hires at CentralU are almost exclusively Drupal, effectively eliminating CSCMS through exclusion. CSCMS does still exist in very small, fragmented pockets, most of which are pursuing Drupal as their next CMS.

CSLMS is currently in review for migration to a new platform. It will still be a centrally supported, single instance, top-down implementation methodology.

Drupal is the unofficial platform of CentralU through sheer volume of developers and sites. There are over 50 developers and site builders at CentralU now making it one of the largest, most active developer bases at the university. CentralU is also a leading voice in Drupal adoption in higher education boasting well over 1000 sites utilizing the platform.

YourNetwork is a recognized part of the LMS ecosystem at CentralU. It still remains decentralized, but has seen increased adoption as time wears on. YourNetwork's development team has grown in recent years as a result of this increased adoption. There is no central support structure planned for this, though mass automation makes this irrelevant. Almost all members utilize CSLMS in some capacity as well but they send students off to YourNetwork for the majority of their learning experience.

Literature Review

Diffusion of Innovations

Complex Adaptive systems and the diffusion of innovations

Rogers (2005) says that small-scale transactions in a system of innovation can often lead to affects at higher levels of an organization. In the case of YourNetwork, a push by one champion of the technology at one department was able to influence that champion's management.

This sets forth a change reaction in an adaptive environment in which those along the managers same social level can now be influenced. Diffusion of innovation models see that these "micro-scale behaviors, frequent instances of adoption, create macro-scale phenomena." (Rogers, 2005, p. 11) The micro behavior of one employee and manager agreeing upon the development of an innovation led to a chain reaction of micro transaction conversations.

A manager tells another manager, they tell their staff, their staff tells other employees and the buzz surrounding the innovation begins to spread like a virus. Rogers illustrates this viral spread of innovation through an S shaped curve.

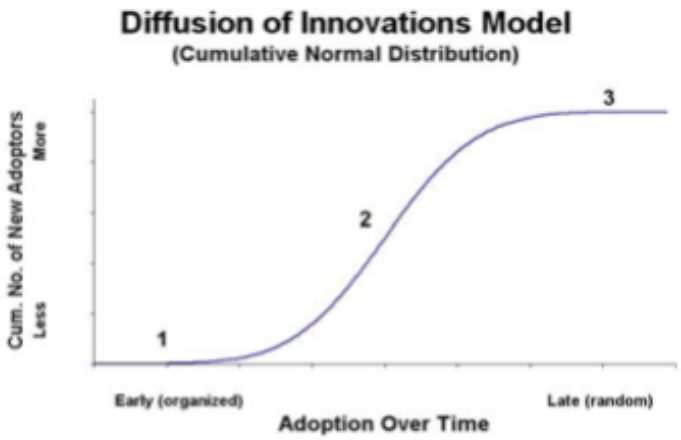

Figure 1 Diffusion of Innovations Model (Rogers, 2005, p. 12)

Initial creation of an innovation is very organized and it's user group small but over time, innovations grow in popularity to almost unsustainable, random levels. The spread of the innovation crosses a critical point (forming the S shape) in which social methods start to spread the innovation at an increasingly rapid pace until, theoretically, all potential users of the innovation have been captured.

Innovation as a Complex Adaptive System

Engler (2009) looks across the spectrum of research about how innovations are born from a process perspective. He groups these process models into five separate generations of innovative product creation. As YourNetwork has undergone a diffusion of innovation, it has also gone through a variety of design phases that can be cross-referenced with each of the innovative process workflows.

YourNetwork was initially formed through experimentation in the *linear model of innovation (Engler, 2009, p. 7)*. It was created without doing any real requirements gathering as an experiment to see if content could be stored local to its unit rather than centrally. This is the starting point of our Diffusion of innovation because it had been created and therefore only affected an extremely limited number of users, those creating it.

After that was met with positive feedback, YourNetwork was redesigned following the *Pull model of innovation (Engler, 2009, p. 9)*. Pull based innovation is where users are listened to ahead of time through requirements gathering. As a result, they shape the initial direction of the innovation and therefore are pulling it toward the product that is to be created. It is at this point in the innovation adoption life cycle that YourNetwork starts to pick up an organized user base and adoption spread to another College.

Engler admits that there is not much consensus on whether generation three and four are different. I would argue they are both slight variations of the same model known as the *Feedback Model of Innovation(Engler, 2009, p. 10)*. The feedback model is an extension of the Pull model in that the innovation of the system is a closed loop. Feedback drives development and development drives more users, which generates more feedback. That feedback is incorporated back into the system once again, completing the loop. This aligns with the beginning of phase two in the motion of the innovation S-curve.

The fifth generation of innovation is when the innovation starts to take on a life of its own. "In this model, networks of innovation are created in which extra-enterprise and cross-discipline organizations join together to innovate." (Engler, 2009, p. 12) Adoption at this phase is seemingly random and the user-base drastically increases.

Open Source

Adopting Open-Source Software Applications in U.S. Higher Education

Open source software projects have increasingly been making inroads in higher education. Williams van Rooij's (2009) review of the literature on reasons for

adoption of open source in higher education found there to be five common threads of rationale. She was able to identify adoption based on the follow reasons:

- Social and Philosophical Believes
- Software Development Methodology Benefits
- Security and risk management benefits
- Software adoption life cycle benefits,
- Total cost of ownership benefits

One critical note of context though is that these perspectives are on the part of technologists instead of educators. YourNetwork has made it a point from its early creation to draw the connection between leveraging open source development and better learning environments for students.

She identified two approaches to software design known as "cathedral" and "bazaar" (van Rooji, 2009). YourNetwork has patterned its implementation directly off of the "bazaar" culture where authorship, resources and management are all decentralized. YourNetwork seeks to follow this approach, which is why I disagree with Williams van Rooij's findings that "looking at open source from the perspective of software development methodology offers little insight about serving both the academic and the administrative needs of an institution of higher education." (van Rooji, 2009, p. 689) Her analysis was conducted in 2008 so we could be on the verge of seeing system implementations begin to mirror development practices.

Some colleges have shown intense interest in YourNetwork, while others appear to have little interest in the project. This could be explained in part by Williams van Rooji's findings about adoption predisposition.

> "The adoption literature recognizes organizational predisposition to open source and the technology skills of staff members as critical to open-source software adoption." (van Rooji, 2009, p. 690)

It would appear that a champion of open source is needed within each College in order for YourNetwork to have any spread potential. The skillset of staff members adopting YourNetwork would suggest that organizations more receptive to Drupal, and in tern have an abundance of that skillset to support, would be more receptive to YourNetwork as a result.

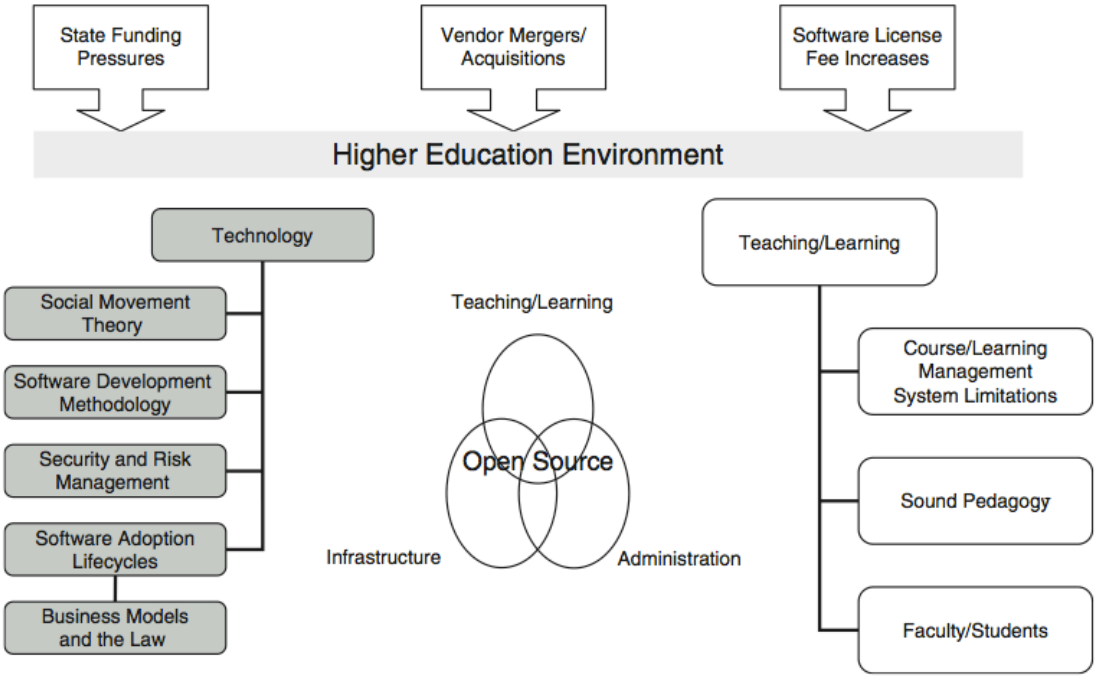

Figure 2 Key drivers of open-source adoption in U.S. Higher Education (van Rooji, 2009, p. 692)

CentralU finds itself, like much of higher education, facing budget cuts. Its central LMS was acquired in a merger and as a result, if it stays with the current solution it will be facing massive licensing fee increases. Williams van Rooij's diagram (2009) about organizational pressures fits the situation at CentralU. Outside market pressures are driving the organization towards new solutions and as a result, employees are turning towards open source.

Organizational Adoption of Open Source: Barriers and Remedies

Some of the many barriers to adoption of YourNetwork are outlined and discussed by Nagy. Nagy (2010) has identified five unique barriers to opens source adoption:

- Knowledge barriers
- Legacy integration
- Forking
- Sunk Costs
- Technological Immaturity

There is a high cost of entry in terms of learning how to use Drupal, the CMS YourNetwork is built on top of. This is where champions of FOSS are critical, as Nagy (2010) suggests that training and outsourcing of system implementations are critical towards overcoming this barrier.

Legacy integration is also a major argument against not just open source development but all new development. Institutions currently work with systems which means new systems would have to interface with the old. Nagy (2010) points to coding and data transmission standards like XML as being the key to overcoming these barriers. Drupal is a popular choice by many institutions as a result of its ability to produce and ingest standards driven data formats.

Sunk costs are related to legacy systems and knowledge barriers. A sunk cost is any resources you have poured into current systems and initiatives. By looking at long-term costs, Nagy (2010) feels that this argument can be mitigated.

Nagy describes forking as "starting with the same source code, if different groups do not coordinate their efforts, the new features and functionality they add may not be interoperable." (Nagy, 2010, p. 150) Forking can lead to multiple versions of the same project or very similar projects being developed. When selecting open source projects it is important to monitor them up front for potential forks in thinking. Often times these situations resolve themselves in time and can be directly related to perceptions of technological immaturity.

Commercial Adoption of Open Source Software

Others were also able to identify four key factors in the adoption of open source technologies in organizations. These had to do with influences towards adoption rather than incentives as others have mentioned. Glynn (2005) groups all negative and positive influences on open source adoption into External Environment, Organizational Context, Individual Factors, and Technological Context.

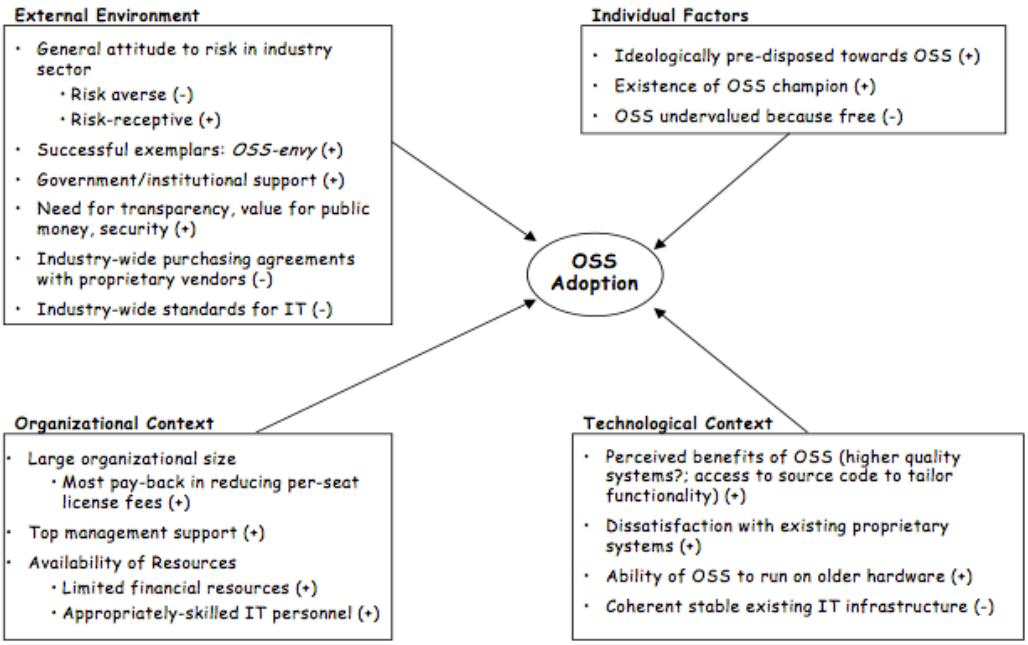

Figure 3: A Framework to Investigate Open Source System Adoption (Glynn, 2005, p. 226)

His framework was able to identify some key influences via interviews. Once again, the importance of having an open source software champion in-house was critical to adoption.

Organizational Technology Decision Making

Factors Affecting the Adoption of Open Source Content Management Systems

Decision makers and influencers of innovation don't always have to be those with the highest-ranking positions. As Lewis' study of Japanese university wide adoption of CMS found, "small-scale adoption is an important way of establishing awareness of and expertise in a particular CMS within the university, which can help to build consensus towards wider adoption." (Lewis, 2008, p. 134) This more organic process of building consensus toward infrastructure decisions are yet another side effect of the open source movement.

It is still possible for there to be top-down decision making within universities but Lewis cautions this because of the politics that can be involved.

> "It is very difficult for university authorities to pronounce that long-established faculty and library websites are to be replaced by a new system, given the traditional autonomy of individual faculties within national universities." (Lewis, 2008, p. 135)

Large universities are traditionally decentralized, with decision-making falling to colleges and departments. Lewis also offers advice to future CMS system implementation within university settings. While "the benefits of a more centralized approach to web publishing are clear" (Lewis, 2008, p. 137), he notes that the social system found within university settings is often times decentralized. This is yet another example where what's best for users from an implementation perspective does not always align with social conventions. Lewis also notes that "enthusiast" (2008) CMS adopters are important but shouldn't necessarily be involved in the final decision.

Using Technology and Constituting Structures Technology

When viewing an organization's use of technology, we must first look at what processes formed that technology. Orlikowski (2000) states that social and political factors remain embedded in technology after it has been created. The people involved in the creation or implementation of a technology have imbued it with certain structural significance that cannot be removed.

She goes a step beyond Winner's "Do Artifacts have Politics?" (1980) though to suggest that initial technology decisions shape and form future social practices.

> "I have sought to augment the existing structurational perspective on technology by proposing a view of technology structures, not as embodied in given technological artifacts, but as enacted by the recurrent social practices of a community of users." (Orlikowski, 2000, p. 421)

Her findings suggest that it is important to focus not just on the politics that implemented a technology selection, but on the ways people utilize and engage with technology as an on-going process. She goes on to identify three types of enactment patterns that users of technology follow called *inertia*, *application,* and *change* enactment. (Orlikowski, 2000)

Inertia is "where users choose to use technology to retain their existing way of doing things" (Orlikowski, 2000, p. 421) and is the lowest barrier to entry and "results in the reinforcement and preservation of the structural status quo." (Orlikowski, 2000, p. 421) This type of enactment is common when there is a lack of connection and engagement with the technology in question.

Application enactment is best characterized by greater adoption of the technology and allowing it to influence the user's workflow. They grow beyond previous norms of how work is to be accomplished and see "noticeable improvements to work processes" (Orlikowski, 2000, p. 422) as a result of adoption.

The third type of enactment she lays out is the most aggressive and is called change enactment. Change enactment occurs when "people choose to use the new technology to substantially alter their existing way of doing things." (Orlikowski, 2000, p. 423) Today the Silicon Valley terminology for this would be one of "Disruption", in which old methods have been completely replaced with new ways of working.

In the case of YourNetwork, we see different groups fitting into all three of these classifications. Two departments, the Founders and a later adopter, would fall into the Change enactment classification. They allowed their way of doing business to be completely transformed and shaped by the technology, which also would suggest that the politics of the technology have potentially had the deepest impact on them. Two other departments would fit into the Application enactment as the technology was ingested, they were excited about it, but it did not yield significant changes in their workflow.

The three remaining adopters would fit into the Inertia enactment category. They felt compelled to utilize the technology because others were, one even suggesting the choice was not theirs to make. In this instance, YourNetwork was a drop-in replacement for old ways of doing business with little fanfare.

Activism

Guerilla Open Access Manifesto

Aaron Swartz of Reddit fame was under indictment for downloading the entirety of JSTOR and releasing it on the Internet. The goal of Swartz was to liberate pay wall based access to scholarly works. He fought for open access to information in a realm that is, still at time of this research, heavily locked down with specialized subscriptions to access knowledge. This mindset is common in Hacktivist culture as many feel "Information is power. But like all power, there are those who want to keep it for themselves." (Swartz, 2008, p. 1)

Swartz feels that activism must be expressed through usage of information technology in order to defeat corrupt systems of control.

> *"It's time to come into the light and, in the grand tradition of civil disobedience, declare our opposition to this private theft of public culture."* (Swartz, 2008, p. 2)

The methods that he suggests be utilized in fighting the corruption of data owners is sum amount to theft.

> "We need to take information, wherever it is stored, make our copies and share them with the world. We need to take stuff that's out of copyright and add it to the archive. We need to buy secret databases and put them on the Web. We need to download scientific journals and upload them to file sharing networks. We need to fight for Guerilla Open Access." (Swartz, 2008, p. 2)

Traditional morality, as defined by the laws of governments, is often askew to hacktivists. Hacktivists view laws as dated and corrupt, leaving the most moral option of new abiding by these laws. "There is no justice in following unjust laws." (Swartz, 2008, p. 2)

Anthropology of an Idea: Hacktivism

While this research does not cover the destructive connotations implied by the term "Hacktivism", it does focus on targeted application of technology. The technology was intentionally deployed in a way that would maximize change within the organization. For this reason, it is a lesser form of Hacktivism and a background on the topic is relevant.

Hacktivism is the combination of Hacking and Activism. McCormick (2013) documents the history of Hacktivism as stemming back to the 50s when early "hackers" from MIT would "phreak" pay phones into getting free calls. The term was coined in 1996 by a hacker named "Omega" in a listserv posting to members of the hacking group Cult of the Dead Cow (McCormick , 2013). This group was an

"explicitly political organization, one that leverages technology to advance human rights and protect the free flow of information." (McCormick , 2013, p. 1)

Today, Hacktivism is often thought of as being synonymous with the hacking activist group Anonymous. Anonymous is famous for Distributed Denial of Service (DDoS) attacks on organizations with whom they disagree with, either for fun or retribution. The following quote by Anonymous summarizes their sentiment well, and is largely what people think of when they think Hacktivism:

> *"We will lay waste to your servers, spread the truth about your illegitimate government, show solidarity with our brothers and sisters in Bahrain, and expose your crimes against humanity" – Anonymous message to Bahrain, March 9, 2012* (McCormick, 2013, p. 2)

Tempered Radicalism and the Politics of Ambivalence and Change

Meyerson and Scully (1995) identify a type of employee within an organization that is at odds with their personal belief systems. While for different reasons than those covered in their original work, Technologist Zero (TZ) from this study fits the organizational anomaly they deem "Tempered Radical".

> *"Tempered Radicals are individuals who identify with and are committed to their organizations and are also committed to a cause, community, or ideology that is fundamentally different from, and possibly at odds with the dominant culture of their organization."* (Meyerson and Scully, 1995, p. 586)

These individuals mean the organizations they operate within no harm; in fact they typically feel compelled to change their organizations to better align with "personal, extra-organizational and political sources of identity." (Meyerson and Scully, 1995, p. 587) This can occur when issues of gender, orientation, race, or politics are in

conflict with the organization. Technologist Zero exhibits political conflicts with CentralU but works to transform the organization from within instead of without.

Voice Lessons: Tempered Radicalism and the Use of Voice and Silence

Douglas Creed (2003) builds upon the work of Meyerson and Scully (1995) by focusing on some of the methods utilized by tempered radicals. Meyerson and Scully (1995) provide specific examples in their study where individuals spoke up to bring about change in the organization. Creed's focus is on how Radicals find their voice, and know when to be silent through a study of gay and lesbian ministers in the Protestant church.

Through this study, Creed (2003) identifies three forms of voice that allow Radicals to drive the message that is needed to remain subversive. The three forms of voice include: the construction of self-hood, voice and silence are ambiguously intertwined, and self-authorization. He suggests that the ambiguity of voice has you "saying one thing, [when] you are not saying another." (Creed, 2003, p. 1528) This allows Tempered Radicals to speak publicly in codified language so those who understand the real agenda to listen.

Still, he found that to members "Outside their [trusted] circle, they do not call it subversion if they speak of it at all." (Creed, 2003, p. 1529) The notion of subversion of the organization was such a powerful thought that it invoked public silence to the Radicals in his study.

Self-authorization is the last component. Radicals tend to legitimize their opinions as in some way being authorized by the organization, even though they know they cannot be spoken. This self-authorization drives them to continue their mission to bring change to their institutions of practice.

> "For the tempered radical, self-authorization may be a critical aspect of resistance that distinguishes enactments of voice and silence designed to make a difference from enactments that border on collusion." (Creed, 2003,

p. 1531)

Knowing what the line is between following the status quo to remain invisible and when it is necessary to voice your politics is critical to enacting change. Too much collusion compromises the Radical's core beliefs, while not enough, potentially exposes their desires to those in power.

Theory

Technology movements and the politics of free / open source software

If physical technology is said to have politics, like a PC, can open source technology in the form of code also have politics? McInerney looks at source code as taking things a step further. "If code is politics, then source code is a prescription for political action, one that can be encoded into organizational form." (McInerney, 2009, p. 210) Open source code openly invites people to modify and participate in the shaping of the technology, an inherently social process. So not only does open source code have politics, open source systems can almost be seen as a vehicle for creating political agendas.

An entire organization can in effect be mobilized by the political intentions of a small group of individuals; McInerney (2009) found this while studying two groups developing for nonprofit communities, one political active and the other not. "When actors attempt to advance their interests by assembling and mobilizing a network, they are leveraging the network to get others to adopt their interpretation." (McInerney, 2009, p. 217) This can lead members at the top of an organization to have even further sweeping influence through imprinting on technology decisions.

Do Artifacts have Politics?

Content Management Systems (CMS) have all reached a point where almost anything is possible. Therefore, decisions about what features will and will not appear in implementations of these systems are inherently political. Much like the blank slate of technology that Winner discusses, decision makers can imprint

technology so that it holds "power and authority in a given setting" (Winner, 1987, p. 10). In any system-wide decision it is then important to know what user groups have been involved in the creation of that system.

Particularly as the size of the organization increases, knowledge of the construction of the technology decreases. This increased specialization affords additional opportunities for those closest to CMS development to have the greatest impact on the political positioning of these systems.

Actor-Network Theory

> "Actor-network theory (ANT) is an integrative approach to science, technology, and society studies (STS) that combines empirical and interpretative methodologies ranging from ethnography to history of science and poststructuralist philosophy." (Tuters, 2011, p. 1)

ANT is relevant to the research study because it describes any transaction between any type of "actant" as relevant to the structure of the entire network.

> *"The complex networks are defined by constantly changing relations between heterogeneous agents (actants) that attribute agency not only to humans but also nonhumans."* (Mulvaney, 2011, p. 1)

Actants are anything that can be connected to and influence the rest of the network. Actants can be humans, technology, objects, anything written down, essentially anything that exists in the world. Latour notes the importance of non-human actants in any interaction network.

> "We know how to describe human relations, we know how to describe mechanisms, we often try to alternate between context and content to talk about the influence of technology on society or vice-versa, but we are not yet expert at weaving together the two resources into an integrated whole. This is unfortunate because whenever we discover

> a stable social relation, it is the introduction of some non-humans that accounts for this relative durability." (Latour, 1991, p. 111)

This reliance on non-human actants to produce "relative durability" is important as it suggests that technologies help human actor networks to achieve the state they ultimately strive for known as Stabilization. Stalder describes the three dynamics that make up actant networks as Emergence, Development and Stabilization. (Stalder, 1997)

Emergence is that "new networks emerge out of already existing ones." (Stalder, 1997) Without actants, there can be no network and so new connections form between existing actors in order to form new networks and behavior. Emergence can also be seen through the growth and expansion of what an actant in an existing network does. "The attempt of an existing actor to grow and include new domains can be a good starting point to observe the emergence of a network." (Stalder, 1997).

Development is that networks can go in two different directions leading to the convergence of divergence of actors from other networks. (Stalder, 1997) This is the interplay of schools of thoughts in which people and actants align with new concepts and modify existing. Actors in the network shape each other though as new networks can deeper connect or disconnect existing ones.

Stabilization is the desire for networks to reach stability, in which actors largely are unchanged. "An actor-network thrives for stabilization because none of the entities which make it up would exist without that network in that form." (Stalder, 1997) Human actors seek stability of the network so that they can form new networks.

Machine to machine relationships, while often created by humans, are equally important because they transform the nature of the human relationships. For example, if a programmer makes two machines talk to each other to automatically

populate data from one system to another, this has eliminated the need for those doing data entry to be part of the equation. Similarly, workers stressed with too much information can better focus on the task at hand when a machine filters spam from their inbox on a daily basis.

It is here in the production of technology as actors that influence larger networks of thought that we are interested in Actor-Network Theory.

Cross-section

Political artifacts and Actor-Network theory can find a crossroads in much of the technology studied in this research. The two meet in an emergent information technology field known as Developer Operations (DevOps). DevOps is an emergent field based in automation. As such, there is still very little writing about the topic and what it means.

> *"DevOps (a clipped compound of "development" and "operations") is a software development method that stresses communication, collaboration, integration, automation, and measurement of cooperation between software developers and other information-technology (IT) professionals."* (DevOps, 2015, p. 1)

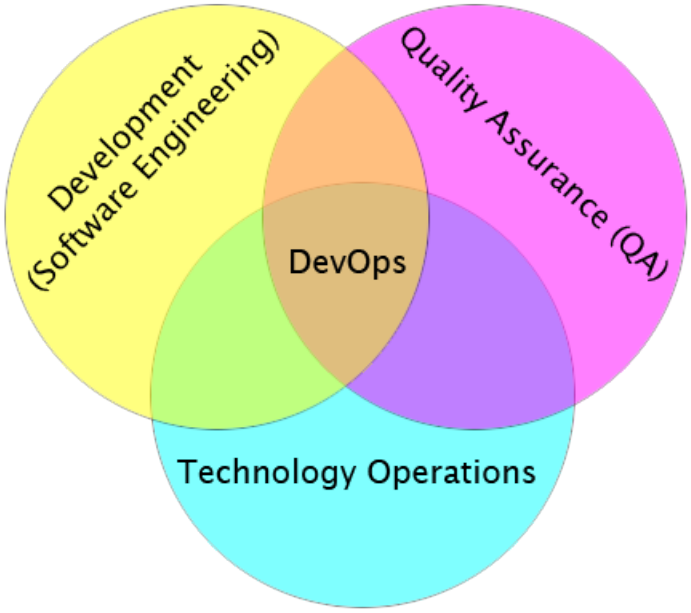

Figure 4 DevOps Intersection visual (Rajiv.Pant, 2012)

DevOps is the capturing of traditional system administrator commands as code. By translating developer process to abstracted code, developers scale themselves far beyond what was previously possible. Wettinger says the major advantage of DevOps is that it "enables fast and frequent releases in the sense of continuously delivering new iterations of a particular application." (Wettinger, 2015, p. 1)

Schaefer, Reichenbach and Frey note the popularity of DevOps in higher education because of its context.

> "Ensuring a high quality of service is mandatory for both, teaching and research, but many institutes find themselves hard-pressed for manpower. Thus, the solution to the plethora of challenges presented by this setting is not to work harder, but smarter. Working smarter almost always means that work is automated." (Schaefer, Reichenbach and Frey, 2013, p. 345)

"Working smarter" turns human to human interaction graphs into simplified human to machine transactions which then perform many complex machine to machine

transactions. This fits well with ANT, which charts all interactions in a system. An example of one such DevOps workflow can be seen in Figure 5.

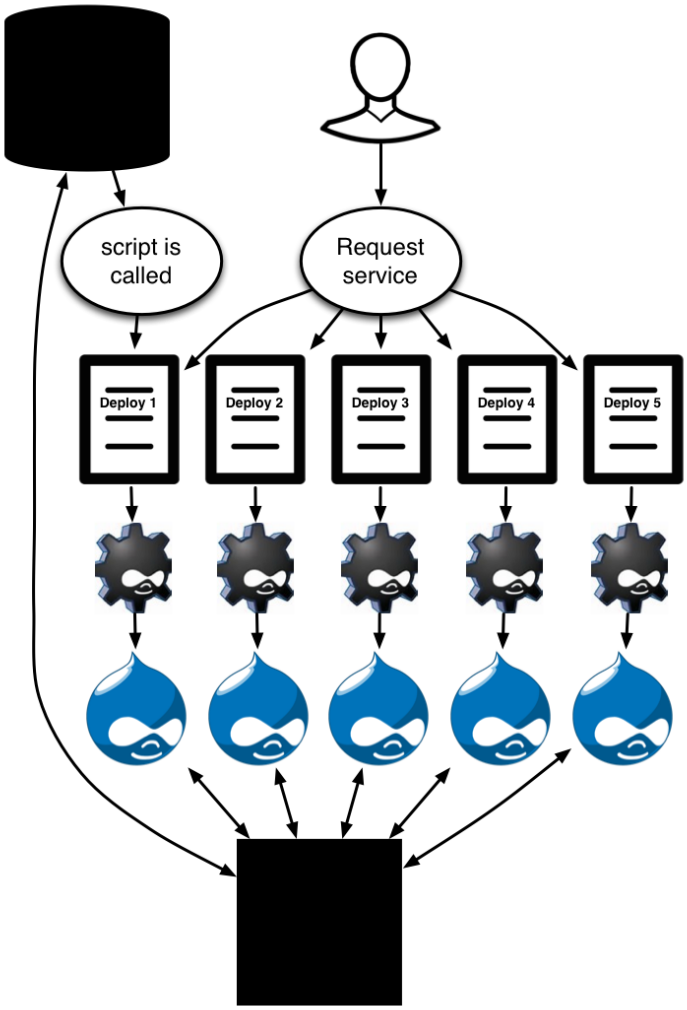

Figure 5 DevOps Workflow

In the figure above, a developer bundles up the operations normally typed into a command prompt. These commands are then replicated and executed five times with perfect accuracy. This is an example process that would normally have required many system administrators, web developers, and managers to initiate.

DevOps is an important cross-section of theories as a politically motivated technologist can scale their influence beyond what was possible just a few years ago. This acceleration of acceleration in technology, and the technologist's ability to

manipulate technology can be seen in many of the services popular today. Facebook, Amazon, Uber, AirBnB, DigitalOcean and others are handling millions of ANT relevant transactions on a daily basis with minimal staffing as a result of DevOps. These businesses all have clearly adopted Orlikowski's *change enactment* (2000), allowing all processes to be transformed by the emergence of new technology practices.

Tempered Radicals utilizing FOSS DevOps workflows can effectively grant technology the perception of agency. By imbuing technology with political philosophies, and loading these into DevOps workflows, technologists can remap human relationships within digital industries. Single actors can orchestrate that which massive corporations once undertook because of this empowering of the machine.

Methods

The research methods employed in this study were chosen to match the social nature of this technologically based research. This is an Autoethnography in which I am highlighting my own efforts to diffuse innovation by making the culture more receptive to the platform this innovation was produced on. Helping facilitate cultural change is a form of social engineering in order to bring this about. Interviews and textual analysis of electronic communications served as data collection methods throughout.

Duration

This research study was conducted over the course of three years. Interviews were performed in year two of the research after Drupal adoption had been increased. This time was chosen intentionally to measure impressions of the YourNetwork platform after the landscape was potentially receptive to it.

Autoethnography

This research is written in my own voice because it is in part the story I lived during my time working with CentralU and it's Drupal community. I wanted to see if we could improve the culture surrounding technology solutions and form community because of efforts I had seen in the larger Drupal community. This community was one of building and collaboration, something that often happened at CentralU in all areas except information technology because platforms were fragmented.

I utilized my passion for building community, socio-technical theory from schooling and ability to code to produce an ecosystem worth studying. My methods for improving the collaborative technical landscape at CentralU involved collaboration both with local CentralU employees as well as other institutions through the larger Drupal community. I publicized everything I was building and helped build CentralU

into a recognized "brand" for Drupal development. In the sustained donation of efforts during the research I was able to identify members to interview, by being a participant observer in the research.

Participant Observer

A large portion of the study involved actively monitoring the spread and success of the YourNetwork platform. YourNetwork did not start as a research project but was an obvious choice of study due to my work in the Drupal community. After a year of production, it was apparent that there was something worth studying and research took over at that point.

Methods used as part of the research involved free release of labor to other departments at CentralU. Sharing of code and ideas with other departments as well as utilization of the Drupal platform to prototype and generate response from other potential adopters. Because YourNetwork is built on Drupal, it was a relevant research method to build and foster a local Drupal community in order to see the affects on adoption of YourNetwork.

By creating a community of experts with knowledge of how the underlying technology to YourNetwork worked, the intent was to see if this could aid in adoption of the platform. By creating a culture favorable towards Drupal, we could see if YourNetwork increased in popularity as well. The actions performed in this research were a form of Tempered Radicalism as Drupal was not the preferred platform of CentralU.

Additional methods in the vein of organizational activism included the building up of key community members, often known as champions, to have a group that could provide its own form of politics. This politics is very nontraditional though in that the "horse trading" if you will often involved time banks, coworking opportunities, free labor, code, and resource sharing. Through the initiation of coworking and

collaborative efforts, borders to areas traditionally non-Drupal were softened to prime them for adoption.

Once a non-member organization showed signs of being ready for adoption, I would dedicate as much time as possible to building them up. If the movement were able to demonstrate, even in a small window of time that it was able to help new potential members, non-members would be more likely to join. By that time, they would be providing their own resources and the intense time investment would pay off in the form of additional workers for the collective.

The previous tactics can largely be summarized as positivist social engineering. Traditional political tactics were reserved only for the groups where technology had almost no chance of changing. In fact, at times, social engineering used reverse network effects to form a victim in the YourNetwork platform.

By alienating and isolating those developers pushing alternative platforms, it lead them to become increasingly hostile towards Drupal and its supporters. This always came across as negativity, and reduced the likelihood that new developers would join alternative platform development shops. They were goaded into giving off the air of being abrasive.

Other forms of social engineering involved management making more traditional agreements to collaborate on projects or to standardize on the same technology. This was a less effective tactic though traditional politics was instrumental in getting the first group beyond the Founders to utilize the YourNetwork platform.

Research Questions

The donation of effort was the primary method utilized in trying to answer my research questions. Drupal was not a popular platform prior to the research project, so the donation of effort was seen as the only way to alter existing opinions and technology adoptions. Interview questions were then framed around who heard

about YourNetwork where, what they thought of Drupal and the application relative to other applications.

In my role as Technologist Zero, I sought to build the larger base of Drupal adoption through donation of effort. Questions were asked to see if respondents were identifying acceptance of YourNetwork as a result of my efforts in working with their colleagues or just by talking to me directly. I framed my questions around YourNetwork though respondents' knowledge of Drupal caused many conversations to reference it as well.

Interviews

Interviews were conducted with 11 different constituents who had knowledge of or were using the YourNetwork platform at the time of interview. The constituents were a random sampling of directors, instructional designers, and technologists from seven different units across CentralU. Only two interviewees were directly associated with the creation of the YourNetwork platform.

The interview responses were captured via written responses to a standard set of questions in ***Appendix A***. These questions were the baseline of what was asked and participants were encouraged to expand on responses through follow up questioning. Quotations were recorded for anything notable but primary driving factors in collection were:

- Where interviewee heard about YourNetwork
- What their impressions of YourNetwork were, initially and after use if applicable
- How the system was described to them by others
- Any emotion laden response to technology, positive or negative
- Any emotion laden response to organization structure, positive or negative

The make up of the interview respondents can be found in ***Appendix B***. This describes the types of people interviewed and their demographic information. It also briefly covers the demographic of those not interviewed as well as reasons it would not have made sense to interview them.

Email / Message textual analysis

As part of the interview process, Technologist Zero granted the researcher access to their communications with others related to YourNetwork. This totaled 1,366 conversations between Technologist Zero and parties interested in YourNetwork both inside and outside CentralU. These conversations consist of email and chat messages captured between July 18th, 2007 and July 30th, 2015. Messages were considered relevant if they mentioned "YourNetwork" directly.

This data is important because it paints a picture of the impact of different individuals and concepts commonly discussed together when referencing YourNetwork in conversation. This data can help enforce and findings of interviews and paint a clearer picture of the landscape and challenges of implementation of YourNetwork at CentralU.

There were five groupings that all conversations have been lumped into as part of textual analysis. These themes were Departments, Systems, People, Actions, and Feelings. Departments and People help to illustrate who was influential, seeking influence, or seeking to further the platform through participation. Systems help express related concepts, things that people keep mentioning paired together with YourNetwork. This helps illustrate what people are looking for that they would mentally associate with the platform and what the platform could help them accomplish.

Actions and Feelings help to convey the mindset of those speaking about the project and their desire for more information. Both classifications drive at the intent of

those seeking out information about the platform or wanting to adopt it and the way they felt about the system.

Technological Agency

The YourNetwork platform and technologies produced to facilitate its perception are another method of research. Technologist Zero expressed that this technology lead to fundamental changes within the organization. As many Respondents described the technology in a way that would suggest agency, the research mentions it here as a method.

Technologist Zero imbued YourNetwork with agency through appropriate application of automation. Other technologies were kept out of the marketplace of ideas as a result of the automation present in YourNetwork.

DevOps is what gave YourNetwork the perception of agency. It could be interacted with via terminals, told to deploy itself on new systems, automatically secure itself, automatically upgrade itself, back itself up, and so on. As a result, with minimal human intervention, it could largely manage itself once told to exist. This freed up a lot of time for me to play the role of Technologist Zero and employ the primary research method studied.

Data

In combing through the eleven interviews, notes from observations, and 1,366 pieces of communication between Technologist Zero and other constituents, five themes emerged. Interviews were conducted with different members of instructional design units at CentralU. The make up of these units and backstories of the types of Respondents interviewed can be found in Appendix B.

The five main themes were:

1. Approachability of technology
2. Anti-establishment perception
3. Groupthink
4. Size, make up and mentality of adopters
5. Independence and empowerment

Each of the five themes were established through utilization of the three data sources in question. Each theme follows a pattern of interview responses, email communication analysis and then researcher supplied observations.

Technologist Zero

One respondent has been intentionally left out of the themes generated below. Their interview went very differently from the rest because they effectively shaped the landscape studied. They are the lead developer of the YourNetwork platform whom we refer to as Technologist Zero.

Technologist Zero felt that the implementation of technology could be done in such a way that it would "necessarily disrupt the established technology systems on campus". Their methods were influenced by the landscape of CentralU.

> "If CSCMS was to follow a pay-based model to adopt open source and CSLMS was to be around forever dull, unchanging and dated, then we would provide an alternative that could not be stopped. I've done a good job of acting like we're fighting against them. I have no problem with their people, I have most problems with their business practices." – Technologist Zero

By producing a counter to the status quo, Technologist Zero effectively created a landscape of "Founders" and "Anti-founders"; those positioned for and against the adoption of the YourNetwork technology platform. Technologist Zero felt compelled to liberate information, a theme common to those that fully adopt the Open Source mindset.

> "When information flows like a river, the flood will drown out the falsehoods. Sure there will be waves and raging storms tossed up filled with false truths for a while. But the longer that information keeps flowing, the more truth and fact will drown out the lies." - Technologist Zero

Part of this mindset meant that TZ frequently gave away code and labor to other organizations free of charge. CentralU was accustomed to having to pay other units for work as per agreements of CSCMS. What was interesting is that by producing work for free for other units, Technologist Zero found a way around the models of the institution that might otherwise have had traditional politics come into play.

> "At first I did it because it was just being helpful to other groups. I couldn't believe how shocked people were that I'd answer email, the phone, or agree to meet with them to help solve their problems. After a few months, I realized that these groups would come back needing more Drupal based work. Eventually I couldn't sustain it and had to turn them down." – Technologist Zero

TZ noted that something amazing happened at this point. Instead of abandoning the platform they had been given, units that had been given free work turned around and hired employees with skillsets that could maintain the solutions.

> "It's as if I gave them a kitten for free and now they needed to pay to maintain and take care of it. I didn't actually believe they would till it kept happening." – Technologist Zero

Technologist Zero felt this process could be replicated and went on to give away seven more "kittens" to other organizations previously not utilizing Drupal. Without a traditional transfer of funds, there was no perception of competition. TZ explains why they felt compelled to operate in this manner.

> "A few of us realized early on that *CSCMS* wasn't going to fly. No one likes it but they are so politically landlocked and many of them seemingly afraid for their jobs that they say nothing. The only people that were interested in it were management crying out for SOME solution, any solution and the developers that already knew the language it was in. No one else wanted it." – Technologist Zero

Technologist Zero (TZ) gave a voice of the voiceless. No one asked TZ to speak for them, and this was not expressed in any of the interviews; but TZ's voice allowed others to question the status quo and select new technology solutions. In a passing conversation with Technologist Zero's coworkers, they mentioned the perceived ability of TZ "to do and say anything" unpunished. It became obvious to other Respondents that if TZ was not to be punished for producing the YourNetwork platform, they could adopt it. Their silence condoned such behavior within the organization.

Electronic Communications

The passion and dedication of Technologist Zero to achieving the goal of disrupting central solutions can be seen in the way that they talk about the platform to others. Word choice also speaks to this being more about a social movement than a technology. Love is mentioned in 13.3% of all communications relating to the project, not a word that would typically come up in reviewing conversation about a technology. The "future" of the platform and advancement of it is brought up 19.5% of the time. The alignment with "Drupal" (59.3%) is also clear, a solution that would not have traditionally been adopted or discussed by their audience if not for Technologist Zero. These untraditional methods and ways of talking about the platform can be seen influencing the themes extracted below.

Approachability of technology

Interviews

The first key trend expressed by 100% of respondents was the familiarity of the technology was critical in their appreciation of it. The technology patterning off of and improving upon existing technology in the respondent's workflow or past work experiences reduced barriers to entry. The approachability also lead to them being able to express to colleagues what the innovation was and why it was important in terms that everyone could understand.

The technology reminded respondents of the Adobe Dreamweaver product that many were developing with at the time of response. 80% specifically mentioned the use of a static html authoring tool such as Dreamweaver, and often times used the phrase "Dreamweaver-like" when describing YourNetwork. Director 3, when describing the technology to coworkers as why they would want to use this over Dreamweaver said, "you don't want the nightmare of Dreamweaver". The familiarity of Dreamweaver was always described in a context demeaning to Dreamweaver because that was the platform all respondents were previously using.

More technical respondents were drawn to the technology being built on top of the Drupal CMS. Technologist 3 echoed this by saying, "I was incredibly impressed, I was on the fence with do we go with Drupal or CSCMS; but after seeing what we could do with the course stuff there…it blew me away".

Concerns, while expressed at times about adoption of something new, seemed to be minor given the dissatisfaction with the current platforms. One respondent, while hesitant, justified their position by relating a story: "I hear people say 'my boss wants to have someone to call'…but I've never called Apple and never called Microsoft". While others were less concerned about the possible risk because of colleagues past experience with the platform "some in the department are already using Drupal for other websites so it's like peace of mind" – Instructional Designer 1.

Colleagues demonstrating the technology at meetings and promoting it also seemed to foment positive sentiment about the platform. "like me I think people were pretty excited" said Director 2. Instructional Designer 3, upon being given a demo went so far as to say thoughts were "positive, useful, thank god, finally".

A specific piece of functionality also was brought up time and time again in interviews, known as the "Outliner". This tool within YourNetwork appeared to be the critical piece of perceived innovation to all respondents. Technologist 2, who would be involved in implementation, said that "They were screaming for it after the demo, they were like 'how can we do this' … Outliner is what really hooked them". Director 3, echoed this was well in reference to the capabilities provided by Outliner "ease and simplicity of the layout, I like that things are drag and drop, move pages around". Another technical decision maker who would be tasked with implementation in their organization said they described the platform to co-workers as "having this unified structure to your courses and being able to use that Outliner to lay things out more organically".

Electronic Communications

In the course of communications about YourNetwork, other systems were brought up. In contrast to respondent interviews, Dreamweaver / "DW" was only referenced 14 times for only 1% of all communications, contrasted with the 80% in interviews. Other systems did come up quite frequently though in terms of comparison or being complementary in nature.

Outliner was brought up in 8% of all communications and could explain why Dreamweaver was mentioned less as this is a newer idea to replace it. While 8% is a small percentage of all conversation, it is relevant as Outliner is but one small piece of the YourNetwork ecosystem and was the only piece mentioned by name.

CSLMS, which would work complementary to YourNetwork, was mentioned in 185 messages for 13.5% of all communications. Similarly, "LMS" came up 10.5% of the time while "CMS" and other CMS solutions were mentioned 11% of the time. It seems communications focused on CMS solutions just as much as they did the CMS's ability to complement LMS technologies.

Observations

Respondents were at ease with the solutions in place, in part it appeared because of both past experiences with Drupal as well as coworkers' experiences with the platform. This is interesting because Drupal was not actually the established platform of CentralU. It appears that the seeding of the landscape by Technologist Zero was effective in making people more receptive to Drupal. Even when respondents expressed concern about maintainability, they still felt both excited and comforted that it was familiar to them.

This conflicting response pattern makes sense given that historically, free open source was not seen as a viable solution within CentralU and had just recently become the norm.

Anti-establishment perception

The platform benefited from not being produced by the established powers that be. A desire for counter-culture as well as centrally supported solutions with high negatives, both lead to respondents feeling overwhelming positive about YourNetwork coming from outside the norm. Institutional pressures to utilize provided solutions, and a lack of pressure to adopt YourNetwork, were also common across respondents.

Instructional Designer 1 said a big reason they went with YourNetwork was a lack of centralization and the fact that there was no pressure from the community. Directors 1 and 2 were also not fans of established technologies, referencing the file limitations of those supplying central technologies.

Instructional Designer 2 felt that YourNetwork was "more energetic and timely; it just seemed better" than central options. They also expressed anxiety over using the centrally supported CSCMS solution. Relative to central solutions, YourNetwork was viewed positively because "it's a very open environment now; it isn't secretive". While nothing was found to suggest a secretive process with central systems, the "us" vs. "them" and solutions from afar appear to have attributed to this feeling.

Questions of strategic positioning also demonstrated a distrust of established technological authorities. Instructional Designer 3 questioned the strategic positioning of IT units having to support YourNetwork and a rival CMS because of the "25k" required to invest in support of CSCMS. They questioned, "if you're going to go off and use Drupal, then why are you using that?" Director 3 also questioned the nature of the relationship with CSCMS because of their local IT group's experience with Drupal.

The perception of local maintainability of YourNetwork also lead to respondents of varying technical backgrounds being more comfortable with it than centralized

options. Instructional Designer 4 felt that "it's fixable" and liked that there was no pressure to use the platform. Technologist 2 was comfortable running YourNetwork and FOSS because they "*never called Apple [or] Microsoft".*

Perceptions of the centrally provided solutions were so negative that Director 1 was surprised to hear other central authorities were supportive of what YourNetwork was providing adopters. The notion of their support in this matter was so foreign they recalled the meeting vividly "where they fully supported [the] concept of content separate from CSLMS".

The separation of powers and philosophy are mixed throughout the other themes. "Keeping content separate from the LMS… in its own place where we could control it" was important to Director 1. They noted that to quell the politics of doing so required a change in speech. By making this "a philosophic change instead of a technical one… that helped build the argument". By establishing a core philosophy to justify adoption of YourNetwork, perceived barriers were softened.

Electronic Communications

Excluding the Founders' responses, we see that while all parties were excited to be doing their own thing, they still wanted to know central authorities didn't have a problem with their activities. 19.6% of all communications were with members of five of the other adopting units. We also see that the first two adopters make up 12% of all conversations, highlighting their enthusiasm in the platform from initially finding out about it. Units supporting CSCMS and CSLMS make up 5.6% of all conversations, showing they are aware but nowhere near as interested as early adopters.

From a user-base perspective, 3.1% of all communications involved those in charge of central units. By contrast, 15% of all communications about the project involved the directors of decentralized units. Taking out the Director of the founding unit, there's still 5.8% of all communications involving the directors of three other units.

This highlights the fervor for this technology solution from the smaller units that would be tasked with going it alone vs. the overall lack of interest from central authorities.

Observations

The anti-establishment sentiment was palpable from interviews. While no one was aggressive towards central units, it was obvious there was resentment from years of being told what to use. YourNetwork appears to have provided a safe outlet for creative expression. While not mentioned as a solution to the status quo, that feeling is there. What's unsaid is as important as what is in this area.

It is not clear if this was always there or if Technologist Zero's influence is seen throughout in this area of the research. While TZ expressed desires of flipping the relationship to one where local control was valued, and those values appear to be present in the local units, the research does not want to suggest causation.

TZ used the phrase "waging an invisible war for autonomy" and this sentiment is definitely there. No one ever expressed the radical behaviors or feelings that TZ did, suggesting that they were potentially unknowing combatants in this "invisible war" for platform. Drupal has a history of social activism and this strain could be seen in the Drupal community at CentralU as a result.

By spreading the Drupal community and Drupal usage throughout all the units in question, they appear to have been ripened for an anti-established mindset. All groups would have been just recently adopting Drupal, something that all parties either were aware was not the same as CSCMS or didn't care. The feeling of being able to run things on their own without central support would easily be explained by looking to co-workers who were using Drupal in their office. This built the sense that they were already disobeying the desires of CSCMS, and as a result they could dissent in other areas as well.

YourNetwork is talked about as if it is either its own decentralized unit or stand-alone technology infrastructure that people go to. While they are all using their own versions of the system in different ways, users still talked as if they were utilizing the same platform. This informal agreement between units to utilize the same technology stack is something strongly latched onto by those involved and flows well into the theme of Groupthink.

Groupthink

The perception of a community of users adopting locally minimized concerns in those interviewed, regardless of skill set.

Instructional Designer 1 worked in a context where Drupal was not the preferred platform, but because some others had been playing with it, this eased concerns. They referenced having watched tutorial videos produced by the YourNetwork founders and Technologist Zero. "I was amazed how well it could be integrated into e-learning and what we wanted to do" from watching those tutorials.

Still, Instructional Designer 1 did admit that it was much harder to have to support your own solution locally. "I had a nervous breakdown one week" they said, in reference to trying to get Drupal setup to their exact specifications. "I was intimidated by it… it took two days [to get past that phase]."

Even in admitting that there was a steep learning curve and that it was a frustrating process, Instructional Designer 1 still felt this was the best idea to pursue. The adoption of Drupal by others in their unit eased their concerns. They said "some in the department are already using Drupal for other websites so it's like peace of mind".

Director 1 echoed Instructional Designer 1's concerns about technical requirements. There is a "perception that you need a technical resource to run it". Through

community though, the respondent believed that the YourNetwork platform "has evolved into something much greater than it was at the conception."

Instructional Designer 2 expressed anxiety about using the centrally supported CMS because of the community's mindset. "I knew it was better to have a community that was supportive". Even though they worked in a very small unit, two people at the time of interview, they were convinced to adopt YourNetwork as a result of where others were looking.

The community surrounding Drupal made them feel like they would be given a better product. Engagement with the Drupal community at CentralU made Instructional Designer 2 feel that "it's more energetic and timely, it just seemed better". They also expressed an alleviation of concerns as a result of past success as by two early adopters of YourNetwork.

Technologist 1 explored how other colleges were utilizing Drupal before ultimately deciding to adopt the platform. While they initially had concerns about "sorta building it from the ground up…" this ultimately drove them to adopt the YourNetwork platform to avoid having to build from scratch. The respondent, who was part of a small unit of two staff members, took comfort in being able to build on top of YourNetwork. "It wasn't like we were completely on our own" said Technologist 1. The respondent's past experiences with CMS technologies at other employers also helped inform the decision to ultimately adopt Drupal and YourNetwork.

Technologist 2 also echoed this sentiment, though they were part of a larger support structure. They had no anxiety about investing energy in adoption "because it was already being used by [Transformational Learning Services] and [they] had preliminary success, I wasn't worried about it". Their concern was more with the need to potentially host the solution, but services provided to their group alleviated this concern.

The technologist also appeared to be comfortable with adoption of new innovations, an affordance that made adoption easier.

"I hear people say 'my boss wants to have someone to call'…but I've never called Apple and never called Microsoft. With every new, different piece of software you put in the enterprise there are support issues but I didn't worry about it more than I'd worry about anything else" – Technologist 2

This past experience with other new and different solutions that ended positively reduced potential anxiety affiliated with supporting an application of YourNetwork's complexity. The technologist's experience with big-box products like Apple and Microsoft, also lead them to believe that even when there was something wrong they still had no real support.

Where anxiety did reside with Technologist 2 came in the form of pace of innovation. It was difficult to discern the best solution because "there's often more than one way to do something. That can sometimes lead to … frustration…you'll often get 3 or 4 different ways to do something." The pace and breadth of possible solutions to a problem is interesting to note because its what drove local solutions to be chosen in the first place. Still, the notion of connecting with peers to solve problems that would arise counteracted any concerns about how to navigate solutions within the platform.

Instructional Designer 3 questioned why anyone would use CSCMS if a community of users were adopting Drupal. When talking to others about supporting the CSCMS support structure as well as Drupal, "[with large dollar amount] invested in [CSCMS], if you're going to go off and use Drupal then why are you using [CSCMS]".

Director 3 and Instructional Designer 4 felt that while it wasn't centrally supported, Drupal was the way to go. "it's fixable" said Instructional Designer 4. Director 3

indicated that they felt their "IT group seemed more in favor of Drupal than [CSCMS]". This is interesting to note because their IT group was supporting both technologies at the time and had steered them toward the newer community technology over previously established solutions.

Director 3 did express concerns about following the community though. The respondent wanted to use what the rest of the community was adopting even though "I was concerned about support…but we needed a solution". They also expressed that there is concern about support regardless of solution.

Technologists appeared to be able to justify following the community based on past experiences with other solutions in the CMS space. The innovation was not a concern to them. All Instructional Designers and Directors expressed some notion of concern about support.

Electronic Communications

The Drupal community and groupthink are reflected in emails as well. 45% of all conversations referenced either "community" or "group". The alignment with Drupal and the larger Drupal community is also apparent. 59.4% of all communications included discussion of Drupal, suggesting that the YourNetwork platform is almost synonymous with Drupal in the eyes of those discussing it. This helps explain why the Drupal community was so influential in groups adopting YourNetwork. People requesting information about the platform are always interested in joining the community by tapping into an existing one that they already knew members of.

The ideals associated with open source and community also can be reflected in communications. Asking for and figuring out how to get help can be seen in 34% of all communications. Meetings and request for more info abound as well. "Meeting" (20%), "presentation" (14.7%) and "demo" (13.6%) come up in conversation

frequently. This speaks to the sheer volume of time Technologist Zero invested in getting people up to speed with the platform.

Open source is mentioned in 10% of all communications as well, highlighting the importance of the solution being open.

Observations

Open source code and culture appears to have been able to change the perception of where technology should come from. CSCMS was open source, but perception of it was negative as a result of the top down deployment utilized. It also had high negatives for being a solution brought to the groups instead of an agreement that they all came to. YourNetwork felt like a community decision that they made together and took ownership as a result.

Technologist Zero and the local Drupal community's work in evangelizing the platform and its ideals could also explain why people wanted to join a movement rather than just adopt a solution. CSCMS was already there, used by 90% of the colleges in some fashion plus central units, open source, and stable. Its capabilities were not that much dissimilar from Drupal, yet Drupal disrupted it. A social movement and charity of resources are the only way to explain people abandoning CSCMS in droves after the learning design community started to align with YourNetwork. People started to follow the methodology of the new organizational structure, distributed, and more importantly free, soured their perception of the old, financially supported support structures.

Activism in the Drupal community also appears to influence the perception of group adoption. The Drupal community would evangelize about the platform, help others, speak about it in an excited manner and get other people excited about it as a result. By the time conversations about YourNetwork were happening, people already knew what Drupal was and were excited to be exploring the use of something else built on it.

These initial experiences had to be positive for early impressions of the platform to be positive as a result. Drupal community members spent countless hours helping new organizations adopt a platform they would have no way of investing in otherwise. It appears to be because of these initially positive experiences that respondents largely ignored their apprehensions about adoption and went ahead fully with platform adoption.

Independence and empowerment

Director, Instructional Designer, and Technologist respondents all used the words "empower" or "empowerment" to describe the system. Director 1 expressed the initial goal was to empower instructional designers, and that faculty also were in the process.

"[We] empowered instructional designers, we didn't really think about faculty empowerment at that time." – Director 1

Director 2 echoed the empowerment of faculty via the platform, and its alignment with their mission.

"We like to teach our faculty to fish" – Director 2

Director 2 also noted the transformation and empowerment of the faculty and instructional designer relationship as a result of the technology. They felt that "it's changed my role as a learning designer" by allowing them to "really focus on learning design and not on look and feel".

Director 3 unfortunately met resistance when attempting to implement the freedoms of YourNetwork with their local IT group. While still appreciating that the

system was easy, they lamented that a feature of automation that saved weeks of development time for other groups, was not allowed by policy.

"I like that it's easy, I wish that we could do the duplicate course feature" – Director 3

In this theme of empowerment, there was a perception of too much empowerment on the part of those doing the empowering. Fortunately this notion is an outlier but worth noting as it could be related to resistance to support local solutions from other central authorities.

Instructional Designers and Technologists echoed the sentiments of management as well. They mostly focused on faculty empowerment that the platform allowed. Instructional Designer 1 did not outright mention faculty empowerment but used the word "empowerment" to describe what they liked about the platform. Instructional Designer 2 expressed that faculty empowered with editing rights "love it". In reference to establishing their access and giving them the ability to work independently they said:

"We'd have these dog and pony shows… we'd check everything off, this is what we've done so far. Now it's at everyone's convenience."

An open, empowering process that everyone was able to do on their own was key to their faculty's positive perception.

Technologist 1 also noted that faculty loved the ability to work independent of the ID unit. "Instructors can build out their own content", which was not possible previously due to technical barriers imposed by existing solutions.

Instructional Designer 4 echoed Director 2's notion that it has transformed the nature of their job, and potentially the whole team.

"It's given us ideas about what our courses could look like and it's given us the ability to experiment with things that otherwise we might not have. And I also think it's going to help open up collaboration with staff." – Instructional Designer 4

The YourNetwork platform's ability to spur discussion and change the units that aligned with it was also expressed by technical staff. Technologist 2 said in reference to how YourNetwork has changed their job, "it certainly has made a lot of things go faster" and "its made a lot of things go away". The ability to modify the platform and control it locally empowered and transformed everyone that it touched.

Electronic Communications

Empowerment was a difficult theme to capture via textual analysis. When grouping together "independence" and "empowerment", it reflected 4% of all communications. This ideal may have been more relevant to those in an interview setting when they had time to collect their thoughts relative to the line of questioning.

Observations

Technical barriers that could be present in local solutions didn't appear to matter to respondents because of the empowering nature of the solution. No matter how complex the system, their stake in a locally supported YourNetwork instance eliminated the traditional barriers associated with "going it alone". They strived for independence to innovate and the empowerment of the users that they serviced. The disconnect between the interview data and the email data set could suggest that the change agent mindset is not present in those requesting information prior to adoption. The bulk of conversations around the platform were request for information as well as getting help and learning about the platform after the fact. The collective influence of Technologist Zero and peer adopters, as well as time to stop and think about it, could have lead the interviews to overwhelmingly express empowerment as being important to them.

Drupal appears to represent empowerment as well to its adopters, which would also suggest this bleeding over into the interview responses. Drupal not being centrally supported necessitated its adopters to empower themselves, both to make the decision as well as to invest in the knowledge required to adopt it. This distributed both the decision making as well as the knowledge about the platform, which in itself is empowering to all those that the platform touches.

The wearing of "Many Hats"

A common theme between all Respondents, regardless of reporting order or job title, was that they all worked in areas that shared a similar state of mind. As YourNetwork was being diffused into units, the size and shape of the shops seems to have been key in why the perception of innovation and emphasis on empowerment were so critical.

Instructional Designer 1 described their group as an "Initiative", which is an informal group or task force typically. The Respondent also identified a common complaint of being someone with "too many hats" as they play the roles of "ID, media specialist, designer, [and] manager of their [YourNetwork] copy". Instructional Designer 2 was part of a do-it-all group of two at the time of interview.

Director 1 was part of a slightly larger group that was a formalized unit consisting of five. The respondent was also a Founding member of the YourNetwork platform, and felt that "Making it a philosophic change instead of a technical one…" was the key to adoption in their unit and others.

Instructional Designer 2 and Technologist 1 worked in small units of two people in size. They both explored what other groups were doing, of similar size, to see what could be utilized. Both described their role as having "many hats". Even units that had more staffing expressed the "many hats" concept though, suggesting it was not only a result of staff quantity.

Director 2, in charge of a group over 10 at the time, referred to their local IT member as having too many hats. Instructional Designer 3 and 4 as well as Director 3 also worked in larger units, but did not express the notion of too many hats. It is worth noting these three respondents also lamented being unable to fully utilize automation capabilities of YourNetwork because of policy.

Observations

Causation cannot be established, but there is a correlation between the units and Respondents that fully adopted YourNetwork's full capabilities. Being stretched too thin as a result of local adoption of technology in other areas could be a cause as well. Receptiveness to build locally could be a larger mindset beyond just that of YourNetwork.

Smaller units, who should have been repulsed by the idea of taking on yet another hat to maintain technology, appear to have reveled in it. They were already used to being empowered to do what was needed to get the job done and this was just one more thing to do. The empowerment theme crosses over into here as well because employees seeing themselves and others as "doing everything" felt that this was the norm. The empowerment seems to have come from other organizational decisions and so these groups would be receptive to taking on "just another hat".

Diffusion of the innovation

Through the interviews, respondents were asked to recall who and how they came to be told about YourNetwork. From these interviews the following chart was created, illustrating how people heard about YourNetwork (see Figure 6), organized based on who the central influencers were.

As all respondents mentioned Technologist Zero throughout the course of the interview, TZ is actually the most connected members of the network. This is to be assumed and our interest with these charts is not to focus on the fact that everyone had talked to TZ at some point, but all the connections between members of the group.

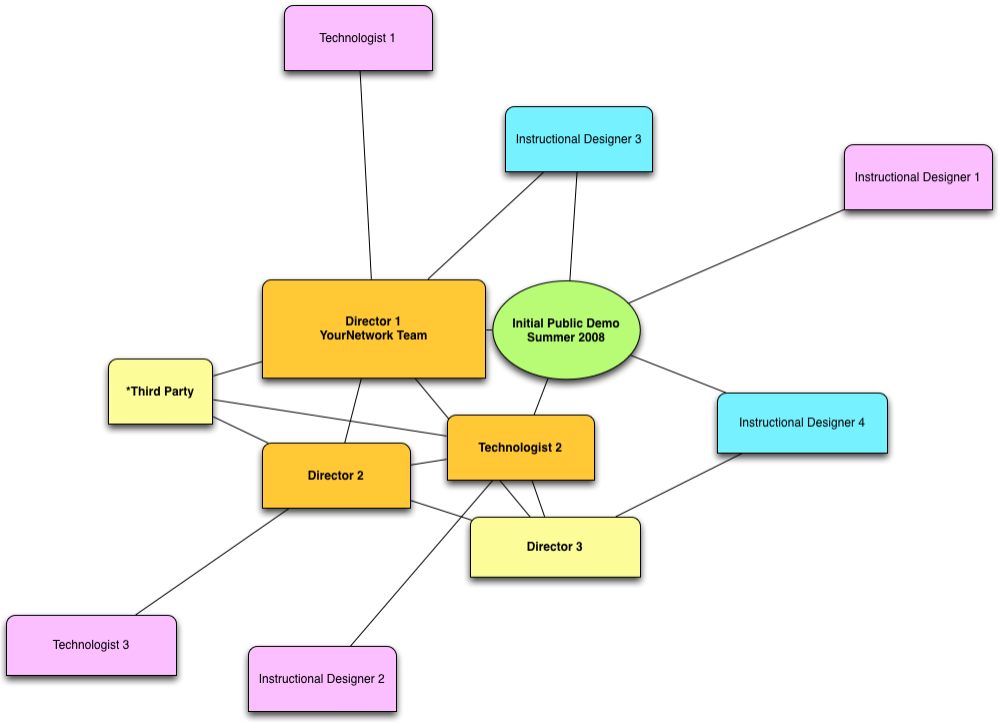

Figure 6: Diffusion of Innovation within CentralU

The connections are organized with those most influential kept to the center of the visual. The color-coding is based on the number of connections to others in the network.

All of the items on Figure 6 are people with the notable exception of the green circle labeled "Initial Public Demo". Many respondents reference this demo, given by the YourNetwork team early in development, as the reason they decided to adopt. It was more than just an adopter to them; the collective experience represented a larger

community of adopters of the technology and could contribute to the Groupthink theme present in interviews.

As we can see from Figure 6, Technologist 2 and Director 2 are almost as connected to the rest of the social network as Director 1, who helped create YourNetwork. This illustrates the importance of champions and passionate early adopters as they provide validation of the initial innovators. Instructional Designer 2 and Technologist 3 cited the organization of Technologist 2 and Director 2 as being the reason that they adopted YourNetwork.

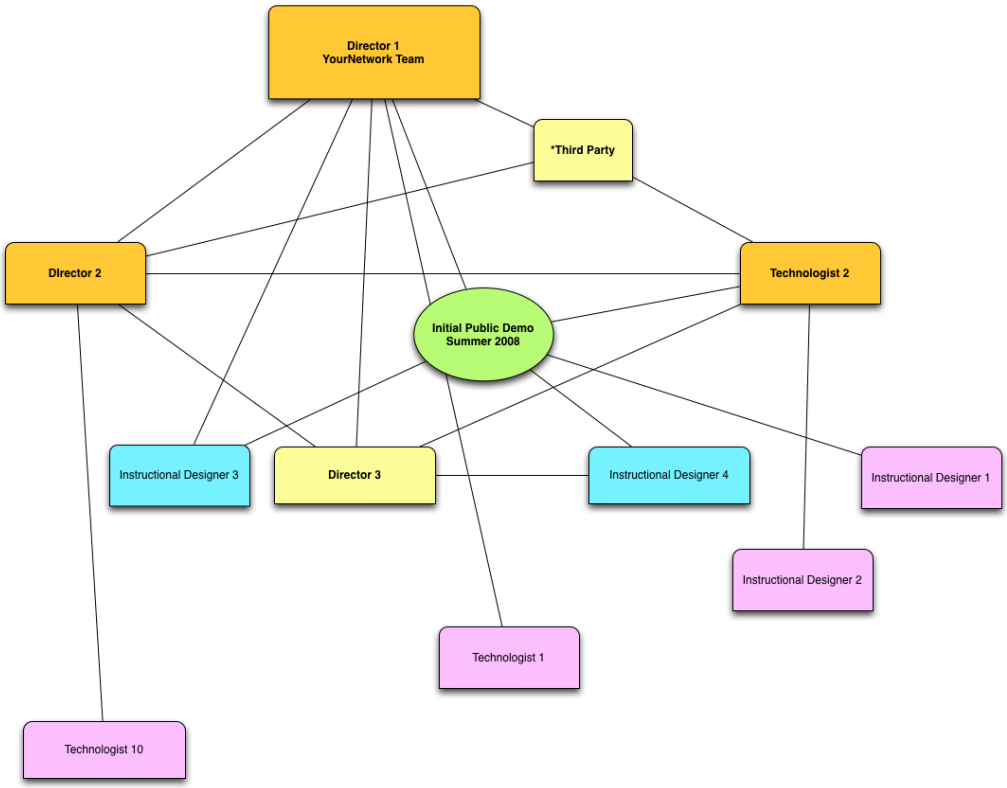

Figure 7: Innovation diffusion based on when people learned about system

The interviews revealed an influential third party that was not interviewed. This third party was critical in influencing Technologist 2 and Director 2 to adopt YourNetwork. The third party was the spouse of Director 1, and frequently worked

part-time with Technologist 2 and Director 2. While Director 1 had discussed YourNetwork with Technologist 2 and Director 2 in the past, the third party appears to have been viewed as an impartial inside voice, leading to the diffusion of the innovation to this group.

Figure 7 places the respondents in order based on when they were told about the innovation. Using this view, we can see that the third party, while only influencing two people, had the largest network effect outside of the authors of the technology. This visual also speaks to the importance that the demo played in gaining additional adopters.

Director 1 and Instructional Designer 2 being excited about the platform going into the demonstration, also appears to have been key in increasing platform adoption. Their fervor at the demo and afterwards helped create the positive community echoed by Instructional Designer 1, 2, and 3. Technologist Zero and the Drupal community's ability to alter the underlying landscape which all worked in to one of Drupal, and all attendees knowing YourNetwork was built on Drupal, made this excitement possible.

Interview Conclusions

YourNetwork's approachability relative to past tools, combined with their acceptance of new ideas and a key piece of perceived innovation lead to positive platform perception. This first time positive impression and approachability trend proved critical to the dissemination of the innovation through the organization.

Technologist Zero's work in seeding the landscape to be receptive to Drupal appears to have paid dividends in the positive perception of YourNetwork. Drupal not being centrally supported doesn't matter to the departments interviewed. They all already knew people using Drupal with success and that was all they needed to utilize the technology. No one was aware that TZ had actually been the one to seed their

departments ahead of time. I am only mentioned in interviews as someone respondents had heard about the YourNetwork platform from. No one mentioned the fact that they had other interactions with me from others projects within their organization. This was not the intent of my actions though it was a very interesting finding. Figure 8 visualizes this fuller picture of the actor network to include Technologist Zero and those that referenced coworkers adopting Drupal.

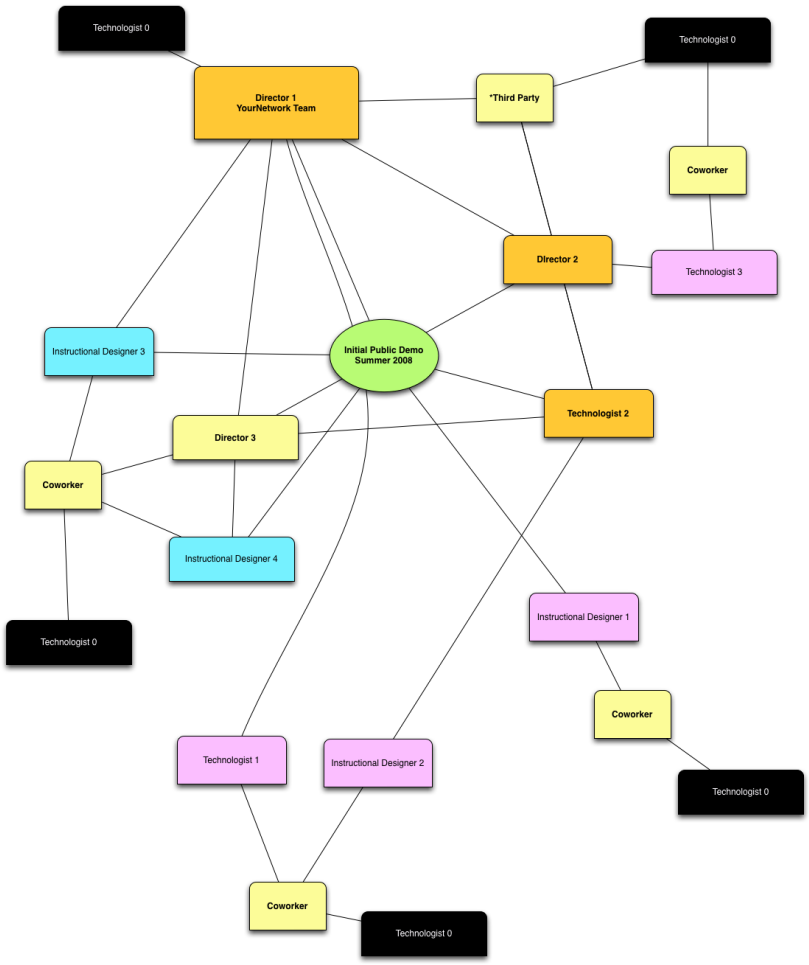

Figure 8 Diffusion of Innovation with Technologist Zero's influence

When adding in these actants, it becomes clearer why people were receptive to Drupal. They already knew other people utilizing the platform in their own work

units. This helped to ease potential concerns of support and taking on too great a risk as there was already a community close to them utilizing Drupal. No one identified, or cared, that I was helping them support the Drupal platform.

Giving away help freely and creating a counter-culture to traditional, financially motivated unit collaboration, appears to have worked to the benefit of YourNetwork as well. The culture of sharing and distribution of knowledge fit with the educational mission of departments. They were accustomed to meeting and sharing pedagogy, making it an easy transition into sharing a common technology solution shaped by pedagogy. Groupthink helped all members feel confident they could leverage each other technologically in addition to pedagogically.

Local control, which would have previously been an issue, was mitigated by the empowering and transformative nature of the tool. The affects of this disruptive innovation can still be seen many years after the original dissemination of YourNetwork, highlighting the lasting impact of empowering others.

Discussion

Information Altruism

The research suggests that there is a force greater than the desires of the organization that drive the adoption and selection of technologies used. It illustrates the power of radicals to transform organizational decision-making using the ideals of Free Open Source Systems (FOSS) as a vehicle.

FOSS code and culture may make people more receptive to what occurred, but it is the act of giving away code, time, help, and knowledge that changed the organization. The free exchange of information technology, counter to financially motivated collaboration, incited change. This suggests a possible tool in the toolkit of organizational activists that I coin Information Altruism.

Information Altruism is defined as the giving of time, resources, and anything in support of new concepts and technologies. This can be used effectively when faced with a social context where money and hierarchy dictate decision-making and collaboration. Through donation of efforts and liberation of information, Information Altruists are able to transform the fabric of organizations through manipulation of the production cost of work. This manipulation of the cost of labor and price paid for information, allows Information Altruists to control the perception of solutions by the output they freely produce, completely devaluing older technologies and business practices.

Socialist notions of equality throughout history come at the cost of hierarchy, infrastructure, and dissemination of those resources equally. Information activism has no need for hierarchy, as technology can take the place of human social

structures. The social hierarchy that emerges is a "do-ocracy", in which there is no formal reporting order but those that produce work are valued the most.

The solution for those without a voice to enact change upon the organization is to start to produce new social structures and norms within those ways of behaving. As an example, Technologist Zero was able to produce a new social norm within CentralU not by attacking CSCMS and demanding change, but instead by building a Drupal community outside of the existing CSCMS community. After identification of those willing to join, information altruism from this smaller community helped grow the community in size.

The donation of effort is completely counter to the previous norms of the organization and momentum built in this new actant network as more and more people joined the Drupal community. The lack of funding and completely decentralized nature of this new actant network led to a realigning of other institutional members with this new way of doing business. CSCMS is made irrelevant not by trying to change CSCMS, but by trying to change individual actants connected to it.

Factors

The factors necessary to identify when and where Information Altruistic methods may be applied successfully include currency, power flow, value system, and flag in the ground. Tempered Radicals can review these factors to apply Information Altruism to their situation.

Currency

There must be a currency, social or otherwise, that is at odds with the Tempered Radical. Whether the currency is monetary in nature or service-based doesn't matter, it just has to be of a transactional nature and something that a counter argument can be constructed against. If financial, then non-financial. If power

consolidation, then power dissemination. For the Drupal community, this was the replacing of financial currency with the non-financial act of effort.

Power flow

There must be a power structure of some kind and a counter must be identified. If power flows top down, the counter is bottom up or middle. The identification of where information and power come from and whom it flows to is critical for identifying weaknesses in the current power flow. For the Drupal community, the counter power flow was from top down to a distributed do-ocracy.

Value System

There must be fundamental principles of the individual that are at odds with the established power flow and currency method combination. Value systems being at odds with organizations are central to identification of Tempered Radicals in the first place. There must be an agitator, champion, and radical in order to facilitate change in the network. For the Drupal community, this was Technologist Zero.

Flag in the ground

There must be a symbol that may be affiliated with the agitators. In ANT, this is known as inscription, leaving things behind for others to react to. Planting a flag in the ground provides a rallying point for those receptive to the ideas of the Tempered Radical to express their deeply held activist ideals. For the Drupal community, there were two flags in the ground. YourNetwork being a technological agent that could produce other technologies created an organic and overwhelming flag in the ground, one that could not be ignored even by those without activist ideals. The second was the formation of the Drupal community. If after seeding the landscape with the technology, there had been no high search results for "Drupal CentralU", then there'd be no social traction.

Other Examples

We can witness aspects of Information Altruism in areas other than just CentralU. Some examples of the transformational nature of Information Altruism in other actant networks follow.

Wikipedia

Decades ago, people had large bookshelves full of vetted and curated knowledge. This was vetted by a small group of people, it was physically produced and labored over and than exported to you in physical, printed form, forever unchanging.

In the 1990's, Encarta and other digital encyclopedias started to change that. They were produced digitally and distributed via CD-ROM. You still had people curating and producing the material and storing it on a physical form, but far less. Information could disseminate cheaper and faster. As a result, less people purchased physical form encyclopedias. This is traditional Capitalism in the information technology space.

But, where Information Altruism is interested is beyond the capabilities of traditional Capitalism to lower costs through efficiency. Wikipedia provides a powerful example of Information Altruism. Wikipedia gives away all knowledge and instead of looking to authority, looks to the collective for knowledge production and curation. It gives away the community curated knowledge and in turn, others do the same.

This alteration of the currency associated with production and dissemination of curated knowledge, in this case money replaced with collective effort, lead to a fundamental shift in the value system of those seeking curated knowledge. The quality is no longer of value because Information Altruism has changed the value system's importance from accuracy to access and cost. As a result, it is no longer

possible to see traditional financial gains in this market and Capitalism is effectively no longer possible.

Mozilla

Another area that we can see Information Altruism at play is in web browser technology. At one time, Microsoft charged for Internet Explorer, as well as many other applications. Netscape, Mozilla, and other early free browsers were initially much lower quality. The issue is that over time though, given that the web has standards and specifications for delivery and display of material, others implemented the specification to greater degrees of accuracy.

Evangelism, especially as can be seen in the Mozilla community, over a long enough period of time effectively forced Microsoft to start giving away Internet Explorer. All new browsers produced are assumed to be free as can be seen with Apple's Safari and Google's Chrome. It is no longer a market that money can be made in selling access to the internet, because enough people banded together and said it was a human right as opposed to something that could be sold access to.

Snowden

Information Altruism is also a tactic for social change more broadly when it comes to the release of sensitive information. The value system in government intelligence is not financially motivated but one of secrecy. The value comes from the networks of people, or lack there of, that have access to information.

The act of releasing top-secret cables by Edward Snowden is a different manifestation of Information Altruism. The reason for doing so, while different from Swartz, is in a similar vein of information activism. Snowden felt compelled to release sensitive information on moral grounds. The information needed to be seen by the public to shine light on the aggressions of governments with respect to a lack of privacy. Snowden's act of releasing these documents to the public planted a flag for discussions of government intrusion and data privacy that otherwise were mum.

Theory Contributions

The research validates the work of Actor-Network theory (ANT) and applies it to a new setting. ANT is used to describe the relationships and influence of humans and non-human actants. It does not express perception of transformation of actants in a network being able to influence the network, as is possible with information technology.

The ability for server and programmable technologies to achieve forms of agency may make it impossible for networks to reach Stabilization. Non-human actants that exhibit growth reinitiate the emergence cycle, preventing it from ever fully reaching stabilization among other actants.

This is different from what many would refer to as "disruption", in which new technologies displace old ways of doing business. Disruption is easy to chart via ANT as it is the emergence of a new network, leading to development of new connections and eventually forming stabilization in a new network.

The autonomous, digital actants in the existing network constantly changes its role in the network through new capabilities. These new capabilities being imprinted onto the digital actants has a direct influence over all existing human actors in the network. These relationships involve the existing actors directly manipulating the digital actants, which allow them to impact and modify the network for everyone else. While this is still the introduction of new concepts and capabilities into the digital actant, it's not actually adding new actants. It is the manipulation and extension of existing digital actants, which prevents stabilization from ever occurring.

For example, you have a phone, email server, and someone that works on the software for the phone and email server. Let's say you go to your phone, hit email,

and it asks the server for email. You do this daily and your interaction flow is linear, always coming to the server to request new information.

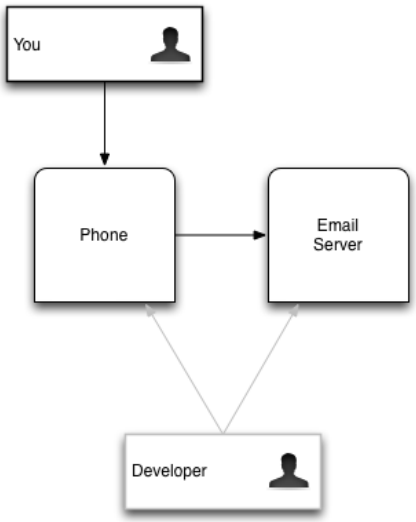

Figure 9 Interaction Flow non-push based email

One day, the developer of the phone and email server decide that you shouldn't have to ask for email, it should be push based. There are no new actants in this network, but the developer has completely reconfigured the digital actants in this interaction flow. Now you no longer push the button to tell the server to give you data; you are constantly being given data at all times, on demand.

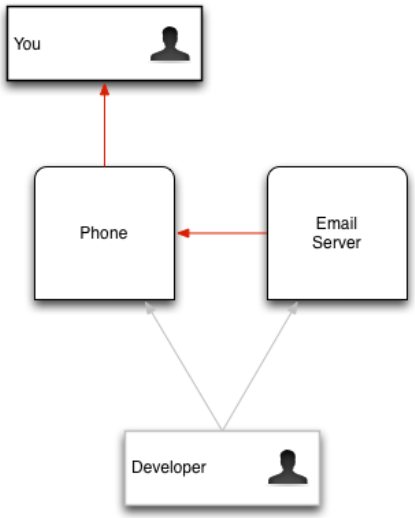

Figure 10 Interaction flow after push based email change

It is this type of remixable relationship that developers can have, imbuing technological actants with new capabilities or modified behaviors. These modifications don't actually add new actants to the network, but they fundamentally change existing actant networks.

Increase the complexity of the relationships and inject activism into the ideals imprinted upon these digital actants and you arrive at the research in question.

Knowledge Contributions

My contributions to knowledge are in the identification of a strain of activism that steams from technology's unstoppable advancement. In the face of misunderstood or corrupt decision making, technological activists seek to reshape the social norms of organizations. This is an extension of the archetype identified by Meyerson and Scully as a *Tempered Radical* (1995) to one that leverages information technology to advance activist agendas.

Meyerson and Scully identified Tempered Radicals working within their organizations to necessitate a change in social norms to better reflect their own ideals. (1995) It could be argued that these Radical's version of morality was in conflict with the belief systems that they worked within. This is relatable to the beliefs held by hacktivists that information and access to it should be free.

Technologist Zero, as identified in the research, was able to carry out the same type of repositioning of morality within CentralU. The shift in ideals from an organization built on currency and authority to one of freedom, community, and local control is a reflection of hacktivists ideals carried out utilizing the methods of a Tempered Radical.

This form of technological radicalism is a significant contribution to the work of Tempered Radicalism via Information Altruism. Information Altruism is an ideal that anyone can adopt as their belief system while the typical case studies where

Tempered Radicalism has been identified revolve around minority and discriminated members of society. This extends the reach of this archetype to encompass any group not currently in majority standing within an organization.

Much of the focus of radicalism is on radical movements in general; engaging societal topics with society at large. These take the form of protests in physical space or the hacking of websites and servers in the digital space. Hacktivist culture is also studied in the context of government or societal level cultural clashes.

Constraint based Activism

This research studies *constraint based activism*, in which actors must try to influence and change the organization while living within its norms. The research contributes a method of behavior that can be followed by activists that want to change their organizations from within. Information Altruism can be seen as altering the social norms of the institution one breadcrumb at a time, all the while staying within the lens of what is deemed acceptable behavior by the institution.

In this way, it contributes to an area of research in activism that is most likely going on everyday within organizations that few just chalk up to "politics". The restrictions placed on the radical to work within the systems they have access to suggests that it is possible for organizations to change from within as opposed to solely needing to be disrupted from ideas from afar.

Information Altruism expresses a methodology for constrained activists of all kinds to utilize internal forces to change behavior of actants. It also positions activist developers as the most capable of invoking change upon the organization, potentially even more so than those from outside.

One interesting constraint in this research is that technology is both the enabler and disabler of action. The activist has to essentially fight technology, and technology policy, with more technology and an alternate, unspoken, technology policy.

Because of this, the context is even more constrained as the only ones able to effectively change technology policy are the technologists that can produce alternate technologies.

Through production of technology, the activist technologist produces a new actant member of the constrained network, which is able to disrupt previously established technologies. Without a new non-human actant, it is not possible to disrupt networks currently witnessed as all human actants have aligned themselves with the technology that exists. Technology creates a context in which, unfortunately, the only solution is more technology.

The imprinting of ideals onto this technology makes the change more acceptable. It becomes the technology that's inciting the change as opposed to the radical actor.

Information Altruism for Organizational Activism

Information Altruism can also be used by radicals to produce information overload in other members of the organization. Radicals can overwhelm the system so-to-speak by producing and disseminating as much information as possible on a topic. By illuminating too much information, real intention can be obfuscated.

When decisions are so complex yet must be distilled quickly by so few, it is easier to influence those decisions when you are tasked with production of that information and analysis. Executors tasked with processing and disseminating information can have more power than those charged with making the decision. Touted as experts because of focus on a field of study, intentional misinterpretation of the facts is even more possible.

This becomes increasingly important as data systems increase in complexity exponentially. The time to assess systems for adoption and implementation is a fixed quantity relative to the size of the organization and number of actors involved. This necessarily will need to change as technology shifts too rapidly to properly

assess and make decisions. It starts to become impossible for actant networks touching new technologies to reach stabilization when the technologies keep change prior to implementation.

Because of this, large organizations will either need to reduce in size and empower smaller teams to make traditionally larger decisions more rapidly, or risk failure. Those assessing the information and selecting platforms that the organization aligns with, have the greatest ability to influence all members. This can be increasingly troubling for those unaware because information technology will continue to allow exponentially more capabilities over time. This means that the degree to which an individual technology decision can reshape and organization multiplies.

A deviation in direction by one degree of a ship heading will lead to being hundreds of miles off course over a long enough timeline unnoticed. As such, it is the individuals making the selection of information technology platforms that sink their organizations or enable them to successfully navigate future waters. By localizing decision making and empowering those directly affected by technology platforms to chose what makes sense for them, organizations can help ensure that radical actors don't steer everyone off a cliff.

The question becomes for radical technologists within these organizations -- what drives you and where do you want to steer your organization? If motivations are lower cost, increased access, and decreased bureaucracy, technology decisions that empower others can be of utmost importance. If the motivation is control, power, and social capital, than it is equally easy for a radical actant to influence technology decisions to benefit themselves more than their peers.

Information Altruism for Managers

Information Altruism can be combatted by management within organizations disrupting themselves before radicals do. For example, creating a culture that

encourages the bubbling up of ideas and management steering as opposed to directing can help prevent radical actants from appearing.

By allowing ideas to bubble up, you ensure that radicals are not at odds with the decisions made as they were part of the process. It is always possible for radicals to emerge so there are two ways of handling them. Eliminate them from the organization (which is obviously not ideal), or cultivate their passion for change and help direct it in ways that are positive for the organization.

Information Altruism for Governments

Governments need to adopt principles of information altruism and allow their citizenry to disrupt dated practices, or risk overthrow. We've seen how the liberation of information in the Middle East led to the overthrow of dictators. Voices previously rejected or stamped out were allowed to flourish for the first time and unite via information technology. Because these governments actively clashed with supporters, they no longer exist. Those previously radicals within the government, as far as desiring freedom and change, now run new governments or have helped instill principles outside of the previous norm.

Governments that are more technologically developed and densely networked like the US need to seriously consider transformation from within or they will be transformed from the outside. They are quickly heading toward a world, because of information technology and a demand for access to information, where nothing is private and nothing can be constrained. Transparency initiatives are a good start, but will most likely not satisfy the demands of information activists to release anything and everything deemed private.

Information Altruism for IT organizations

This suggests that gatekeeper style information technology management practices are unenforceable. As free options increase in quality and decrease in complexity of deployment, it will become increasingly difficult to enforce policies that require the

usage of centrally mandated anything. This is because of the relative obscurity afforded to those that work in a large enough organization in general. Scope of what all members do becomes muddled because of the overabundance of data and goals of the organization.

As information spreads freely, human beings will desire that style of freedom in every aspect of their lives. This has been seen with the rapid collapse of multiple governments in the Middle East, but also can be witnessed in any large organization with policies that are at odds with the current capabilities of technology. People will find a way to use the tools they want, regardless of policy.

This can be changed if IT organizations adopt an organizational structure of flatter, smaller teams. Silicon Valley companies like Amazon and Google have adopted this style called by some the "two pizza rule" (Brandt, 2011). Following this rule, individual ideas are valued the most and bubble up from small teams. These small project teams are kept to a size where two pizzas could feed all members.

This size organizational structure and bubbling up of the best ideas helps to ensure that radical voices are heard and actively used in decisions made. A collection of voices to form decisions could help dissuade radical actors from emerging. Smaller teams could also help improve transparency and accountability, making it more difficult for radicals to invisibly foment dissent. There's presumably no dissent to foment in the first place as all voices are heard.

Information Altruism for Tempered Radicals

For Tempered Radicals, Information Altruism represents approaches to organizational adoption of new social norms through information technology. By creating technological actors and imbuing them with the ideals of their creators, ownership of ideals can transfer be imprinted onto technology. When structuring

conversations, reference the ideals and philosophies used in production of the technology as opposed to your own.

Often when speaking about the construction of a technology, no one stops to ask who or why it was produced that way. Take a series of hypothetical statements about positioning technology within the organization.

"We roll this out locally because it's been shown to provide better outcomes for users by giving you greater control over what you want to do". This is a heavily codified statement laden with libertarian ideals. Self-empowerment, improved products, improved services, all great things, all positive things, but it could still be questioned why you want someone to run their own technology. The messenger is easily associated with the message, so you are the one wanting them to deploy locally.

These same prospects can be made threatening if the same message is framed in a different manner. "We don't want other people to be able to tell you want to do. They don't know what you really need and what your people want". This communicates the same message about technology as above, but usage of "they" provides an "us vs. them" proposition and is potentially aggressive, even if the ideals are the same.

Abstracting our ideals and imbuing technology with them though, the radical can produce a different kind of conversation about technology. "To run this you just grant us access to a server and our solution will be setup for you". We can provide service for you; there are no ideals and no codified language. It's completely removed from what the original use-case was and the focus is on the technology providing the service. Let's take it a step further now.

Remove the human actor from the equation entirely. There's no human-to-human interaction at all. Create a website. If someone logs in and fills out their information,

your ideals, your efforts, your vision of the world, and how they should interact with technology, is all automatically setup for them. The radical now focuses on conversations that bring them to the conclusion that solutions should be run locally, and the technology is the actant ensuring that it is easy to have happen. Web services are no long people, they are completely dehumanized but, the technology they produce and the work they replace can still be threatening.

The radical is no longer the threat actor in this scenario though; it is their website and web services. With a little information, it does the work all on its own with no direct involvement of the radical technologist. Deployed in an organization that is financially motivated, this free transfer of resources would be too threatening for a human to propose.

Information Altruism suggests that radicals can change the behaviors of the organization to be more receptive to their worldview through the act of giving away information technology. The important point here is that that technology has been imprinted with ideals that are fundamentally at odds with the organizational culture. This non-human actant that has the ability to directly influence the rest of the actor network, can provide the activist a choir of voices even though they established the actant.

Information Altruism for Activists

For activists, this case study should emphasize one thing -- radicals must drop the radical pose to achieve the radical ends. This is an area where Tempered Radicalism and constraint-based activism can be of most value to unconstrained activists. Lasting change must not be threatening to the institutions that you wish to transform or it will be rejected. Constraint based activists have the concern of job loss or worse, where unconstrained activists, most typically seen as hacktivists and demonstrators in the streets, do not fear consequences of action. The problem

though is that often times without constraints, there is no unified plan of action, or the plan of action is not sustainable.

Technology can be a tool to transform institutions and establishments who are not willing. Open Source, and the gifting of time and energy to others are critical in gaining support for your cause, no matter what it is. For those seeking to achieve social change, it is important to couple technology with methodologies that induce empathy, not division. By imbuing your technology with these ideals, you can ensure its adoption more organically increases. Utilization of social media is also important in building arguments. The echo chambers of retweets, likes, favorites, and otherwise all add to the perception of opinion within a network of actors.

Social media and the leaving of data for others to witness allows for the production of many more actants than in the past days of activism. As important as the message activists try to portray is the messaging itself, a lesson that can be learned from the passive aggressive methods of information altruism.

Also, there is something to be said for the perception of organic distributions of innovation and ideas. Supplanting ideas worth spreading and making your ideas bigger than they are can often spark change to occur in the way you are trying to shape it.

Information Altruism builds the argument that it is possible to change organizations from within the constraints they provide. As such, radical actors that wish to change institutions cannot stand outside systems of control and simply protest. They need to embed themselves deeply within them, find each other, and devise ways of transforming institutions from within. To quote Alinsky, “Learn a lesson. Go home, organize, build power, and at the next convention, you be the delegates.” (Alinsky , 1972, p. xxiii) The way to influence and change the system is effectively by becoming one with the organization.

Future research

It is beyond the scope of findings of this research, but it would be interesting to follow the course of society after the extended application of Information Altruism. As we can see at the time of this writing, foundations like Mozilla and Wikipedia now effectively rely on donations to sustain themselves. They are finding that after collapsing Capitalism in these markets that one needs to find a way to sustain them as the economy they work within is still currency based!

For centuries, the representation of time units in the physical form of currency allowed people of different skillsets to trade and benefit from one another. Historically, you could barter for goods and we may be on this track towards a more barter driven society in part because of people devaluing traditional forms of currency through information technology adoption and evangelism.

Paul Mason, economics editor Channel 4 in the UK, suggests that we are witnessing the beginning of the end of Capitalism. In a July 15th, 2015 piece for The Guardian he points to information technology, the desire for information to be free, time banks, crypto currencies, and other examples as signs that the world is in a slow transition to a Postcapitalist realm.

> 'New forms of ownership, new forms of lending, new legal contracts: a whole business subculture has emerged over the past 10 years, which the media has dubbed the "sharing economy". Buzzwords such as the "commons" and "peer-production" are thrown around, but few have bothered to ask what this development means for capitalism itself.' – Paul Mason - The end of capitalism has begun

Information altruism provides another example of how Paul's postcapitalist society can get started. What remains to be seen is where it goes from here and how it sustains itself, or if it even can.

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Appendix A: Interview Questions

- What department do you work for?
- What is your role at the university?
- How did you first hear about the e-Learning Management System, also known as YourNetwork?
- Were you one of the primary decision makers in adopting YourNetwork?
- (if secondary adopter)
    - Was their pressure to use YourNetwork or was it an alternative to other methods you were currently using?
    - Were you presented with other alternatives to YourNetwork? If so what ones?
- (if decision maker)
    - Why did you first begin investigating YourNetwork?
    - What was your initial impression of YourNetwork?
    - Was their pressure from others either to use or not use YourNetwork? If so, how so?
- Did you feel compelled to keep YourNetwork to yourself or share your experiences with others either inside or outside of your department?
    - Was there ever pressure to do either?
- Do you think YourNetwork should become a central service or remain local and distributed?
    - If confusing, do you like that your version of YourNetwork is locally supported by your staff or do you think it should be a central entity like CSLMS?
- What are some of the things you like about YourNetwork?
- What are some things you don't like about YourNetwork?
- What would you like to see change about YourNetwork going forward?

- What do you think YourNetwork has done for you?
    - What has YourNetwork done for your staff?
    - What has YourNetwork done for your college / department?
- Social Creep / Word of Mouth

Appendix B: Interview demographic

Interviews were conducted between March and June of 2010. The questioning was open ended to allow follow up and the interviewee encouraged branching from the base outline to try and elicit better information from the respondent. There were four key demographics interviewed that are worth noting.

Founders

Founders were anyone that was directly involved in the production of the YourNetwork platform from the point of conception in 2007. Founders all worked in the same department which acted as a Skunk Works. The team was small, new, diverse in age and specialized in skillset. The specialization of the Founders allowed them to fit into the other three categories of Respondent.

Director

Directors were all previously Instructional Designers in past careers and appointments. Their background in Instructional Design helped them to understand the importance of being able to shape education technology to needs. All Directors each had over a decade of experience working in the field prior to becoming management. They each ran their own small to medium teams, and met frequently to discuss best practices. All directors had positive interpersonal relationships, which encouraged their departments to meet regularly and share ideas openly.

Instructional Designer

Instructional Designers facilitate the delivery of courses using technology. None of the ID Respondents had a programming background and all held a preference towards utilizing existing technology to solve instructional challenges, rather than building new solutions. YourNetwork represented empowerment to this group because it tends to be underserved and overlooked when it comes to the design of

technology solutions. They took ownership of YourNetwork because they helped directly shape it to shape instruction instead of the other way around.

Technologist

Technologists had all held previous employment in other industries prior to working at CentralU. They typically worked a bit more isolated from the rest of the teams that they were embedded with due to cultural norms associated with IT work. All Technologists had programming experience of varying degrees though all were self-taught with only a few years of on the job experience doing development work at the time of the interview. All of them wore multiple hats when it came to IT work and all worked in relatively small departments at the time of interview.

Adopting Departments

There were two major classifications of departments portrayed in the interviews. The first type required everyone be able to do everything to get the job done, regardless of size of the team or scope of the project. If someone left, the idea was that everyone else could pick up the slack. These organizations are able to produce more output but at an overall lower quality with lower risk of incident in the event of a departure. No member is critical to the success of all projects and all members share responsibilities equally. All duties are as assigned.

The second type of department in the research is generally the same size but the make up of employees is completely the opposite. There are no generalists; everyone is an expert in their piece of the puzzle. Projects are highly collaborative but also all members pulling their weight is required for any project to be successful. As a result, they produce consistently less output but at a consistently higher quality. Projects are higher touch and have a higher risk of failure with the departure of critical team members because responsibilities are distributed based on skillset. All duties are as they match skill set.

Non-adopting departments

Departments not selected for interview due to resistance to adoption also fit into three classifications.

Anti-founders

Anti-founders were advocates of parallel but perceived conflicting technology stacks to those of the founders group. It is not to say they were actively against YourNetwork, they already supported other technologies including CSLMS and CSCMS. As a result, they would not adopt a fundamentally different missioned technology.

Too Many Hats

Groups wearing too many hats appeared to not reach a critical threshold for adoption. The challenges were perceived to be too daunting to accept the responsibilities of adoption because team size. This threshold appeared to be present in groups with three employees or less. All of these units hired generalists out of due to their size and risk of total department failure.

Traditionalists

The Traditionalists worked in historically large, slow moving organizations. These organizations were established for the purposes of scale and stability. As a result, it makes anything new very difficult to ingest and process in their organizations. These organizations had many levels of managerial hierarchy with very large project portfolios. Some also fell into the Anti-founders classification in that they produced their own, parallel platforms to YourNetwork bringing them into a position of perceived competition. These groups are generally the last to adopt innovation and largely sit back and see how innovation disseminates into smaller groups before making decisions of mass adoption because they are, by design, risk adverse.

Appendix C: Conversation Data

1,366 conversations were recorded and analyzed for common themes based on reviewing email communications. These conversations were captured via email and chat and were organic in nature. They took place between July 18th 2007 and July 30th 2015 and included a random distribution of people involved in the discussions.

The Five categories were People, Actions, Emotions, Systems and Departments. These were chosen as they had a significant amount of results when performing a basic textual analysis. These themes also helped to add validity to the interview data which often times involved the same people, organizations, themes and technology conversations.

Analysis consisted of filtering all communication based on the exact mention of "YourNetwork" in the conversation thread. It was then paired with a keyword that fit into one of the five categories and a count of the number of threads matching this cross section was recorded.

Appendix D: IRB Approval Form

The Office for Research Protections (ORP) has reviewed the eSubmission application for your research involving human participants and determined it to be exempt from IRB review. You may begin your research. This study qualifies under the following category(ies):

Category 2: Research involving the use of educational tests (cognitive, diagnostic, aptitude, achievement), survey procedures, interview procedures, or observations of public behavior unless: (i) information obtained is recorded in such a manner that human participants can be identified, directly or through identifiers linked to the participants; **and** (ii) any disclosure of the human participants' responses outside the research could reasonably place the participants at risk of criminal or civil liability or be damaging to the participants' financial standing, employability, or reputation. [45 CFR 46.101(b)(2]